Intraday Jumps and 0DTE Options: Pricing and Hedging Implications

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Abstract

Zero-days-to-expiry (0DTE) options have transformed intraday market dynamics, displaying extreme sensitivity to abrupt underlying price movements. This paper investigates how intraday jumps affect the pricing and hedging of 0DTE options on the S&P 500. We develop a continuous-time stochastic volatility model with Poisson jumps and derive semi-closed-form solutions for European option prices, enabling us to isolate and estimate market-implied risk premia for return diffusion, volatility and jump risks. We find that intraday jumps in the underlying index are both frequent and impactful, with intensity clustering around the market open and close. Using 22,410 five-minute observations of SPXW 0DTE, we document that the implied premium for jump risk is not only statistically significant but nearly twice as large as the combined premia for diffusion and volatility risks. These results suggest that 0DTE options provide a powerful tool for hedging against extreme intraday market movements, offering new insights into how short-maturity derivatives reflect tail risk in real time.

Keywords: ultra short-dated options; diurnal patterns; jump risk premium; extreme price movements; event-based trading

JEL classification: C51, G12, G13

1 Introduction

Trading in zero-day-to-expiry (0DTE) options has gained significant popularity in recent years. Listed one week in advance, these contracts expire at the end of the same trading day. Since Cboe expanded its S&P 500 (SPX) expirations to every weekday in 2022, 0DTEs have accounted for half of all SPX options volume (Xu, 2023). This rapid growth signals a structural shift in how traders express short-term views and manage intraday risk. Unlike longer-dated options, 0DTEs exhibit explosive gamma profiles, in which small changes in the underlying index can trigger large hedging flows from market makers. These flows can amplify intraday volatility and liquidity demands, creating a feedback loop that directly affects price discovery in the underlying index.

Despite their systemic importance, the literature on 0DTEs is still nascent. Notable recent exceptions mostly explore variance risk premia (Almeida, Freire, & Hizmeri, 2024), implied volatility surfaces (Bandi, Fusari, & Renò, 2023), spillovers to the underlying market (Brogaard, Han, & Won, 2023; Dim, Eraker, & Vilkov, 2023), implications for retail investors (Beckmeyer, Branger, & Gayda, 2023) and 0DTE-based trading strategies (Vilkov, 2023). However, understanding the drivers behind the explosive growth of 0DTEs remains a critical open question. Two distinct motives likely explain the rise: cost-effective hedging and speculative trading. On the one hand, 0DTEs enable institutional investors to efficiently target specific intraday risks. As Andersen, Fusari, and Todorov (2017) note, short-maturity options are ideal for hedging tail events without loading heavily on volatility risk or time value. On the other hand, the retail segment appears driven by speculative motives. Agarwal, Ghosh, Prabhala, and Zhao (2025) document that financial constraints and lottery-like preferences drive retail participation toward short-duration directional bets as options converge to expiry. Whether driven by hedging or speculation, the 0DTE market represents a distinct segment with peculiar pricing dynamics. It provides new, valuable

insights into investors' preferences and risk premia over intra-daily horizons.

This paper assesses the impact of sudden, extreme underlying price movements ("jumps") and the associated risk on 0DTE SPX option prices. We focus on jumps for two reasons. First, as Andersen et al. (2017) rightly point out, the short-term options are a natural laboratory for learning about tail risk driven by these extreme movements. Second, jump and diffusive risks are compensated differently (Liu & Pan, 2003), and jump risk premia account for a significant share of the total premium (Santa-Clara & Yan, 2010). Extreme price movements are also linked to market health (Jawadi, Louhichi, & Cheffou, 2015), and research on price jumps facilitates the detection and diagnosis of potential market instability. Evans (2011), Lahaye, Laurent, and Neely (2011) and Xiao, Yin, and Zhao (2020) emphasize the significance of intraday price jumps, documenting that between 33 and 43 percent of them correspond to US macroeconomic news announcements. Lee (2011) finds that, in addition to macroeconomic news, jumps are also often associated with Federal Open Market Committee (FOMC) meetings and companies' earnings announcements. Consistent with this, Jacobs, Ke, and Pan (2026) document that option-implied tail risk measures robustly predict market returns around these FOMC announcements, highlighting the importance of tail expectations in shaping risk premia. Pure macroeconomic or earnings announcement effects cause significant increases in the absolute sizes of jumps, and the informational surprise of the announcement explains large proportions of the jumps. Dim et al. (2023) argue that ultra-short-dated options' high leverage and gamma risk make them good candidates for event-based trading.

Figure 1 illustrates the sensitivity of 0DTEs to such events using market dynamics from January 31, 2024, a day characterized by a pivotal FOMC meeting. As depicted in the top panel, the SPX fell approximately 1.6% in a series of negative jumps following Federal Reserve Chair Jerome Powell's comments, which pushed back against expectations of rate



Figure 1: Intraday dynamics of SPX and a 0DTE put option on January 31, 2024. This figure illustrates the intraday price and volume movements, captured in 5-minute bars, on a day characterized by a significant market response to an FOMC meeting. The top panel shows a candlestick chart of the S&P 500 (SPX) index. The middle panel presents the candlestick chart for an out-of-the-money 0DTE SPX put option with a strike of 4860. The bottom panel shows the corresponding trading volume for this 0DTE put. The data are from Cboe.

cuts. The middle panel focuses on the corresponding price movement of an out-of-the-money 0DTE SPX put option with a strike price of 4860. This option opened at $2.20

and traded at a relatively low premium, followed by a dramatic increase in value, gaining around 640% from the open and around 2400% from its intraday low. This behavior vividly illustrates the potential for significant returns inherent in 0DTE options during periods of extreme intraday market movements and pronounced tail events. The bottom panel shows the corresponding surge in trading volume for this specific 0DTE put, reflecting heightened market activity and interest in either hedging against or speculating on these rapid price fluctuations.

The broad importance of intraday jumps thus highlights the need to examine their pricing and hedging implications on 0DTEs. We investigate whether there is evidence of 0DTEs' intraday jump risk and how this risk affects the implied volatility smiles. Specifically, we compare the magnitude of the intraday jump risk premium implicit in 0DTE prices with the return diffusion and volatility risk premia, and determine whether these instruments act as efficient hedges against extreme intraday movements.

To address these questions, we must account for the unique characteristics of 0DTEs. Their payoffs are determined entirely within a single trading day, making them disproportionately exposed to discontinuous price changes. We employ a general continuous-time stochastic volatility model with time-varying Poisson jump intensity to capture the index's underlying dynamics. The instantaneous variance follows a mean-reverting diffusion given by the "square-root" specification à la Heston (1993). We infer the model parameters by applying the Efficient Method of Moments (EMM) proposed by Gallant and Tauchen (1996) on the 9,164 SPX five-minute bars. The EMM procedure relies on an auxiliary model approximating the conditional distribution of returns. We use the diurnal adjustment from the Autoregressive Conditional Duration model of Engle and Russell (1998) to represent diurnal patterns in intraday data. We then transform the model into its risk-neutral form to derive semi-closed-form solutions for European option prices. This setup

allows us to construct the model-implied volatility surface and infer risk premia using a dataset of 22,410 five-minute observations of SPXW 0DTEs.

We demonstrate that intraday jumps in the underlying index are crucial drivers of the 0DTE price dynamics, exhibiting a stronger pattern than diurnal volatility. The intraday jumps are significant in frequency and magnitude: on average, there are about 4.5 jumps per day, and the average jump size is -2.1 basis points. These metrics capture within-day discontinuities that are distinct from the larger, less frequent daily jumps typically identified in longer-horizon studies (e.g., Andersen, Benzoni, & Lund, 2002; Bakshi, Cao, & Chen, 1997; Bates, 2000; Broadie, Chernov, & Johannes, 2007; Eraker, 2004; Pan, 2002). The distribution of jump magnitudes is left-skewed, with negative jumps occurring about 55% of the time. The average and the standard deviation of jump sizes are also greater for negative than for positive jumps. Jump intensity clusters near market open and close, coinciding with high-volatility periods. Finally, the overall jump risk premia implied by 0DTE prices are significant and exceed the combined premia for return diffusion and volatility risks.

This paper makes several contributions to the option pricing literature. First, to the best of our knowledge, this is the first study to develop an ultra-short-term option-pricing method that incorporates all the prominent features of intraday index dynamics. Second, the proposed model provides a semi-closed-form solution for the prices of ultra-short-term European options. Third, we demonstrate that a bimodal distribution of jump sizes effectively captures extreme intraday index swings, as the behavior of positive and negative jumps is highly asymmetric. Fourth, the jump risk premium is substantial and clearly priced by the 0DTEs, consistent with similar findings of Eraker, Johannes, and Polson (2003) and Bollerslev, Li, and Todorov (2016) for longer-tenor options. Fifth, we show how 0DTEs can be used for hedging extreme intraday market movements.

The remainder of the paper is organized as follows. In Section 2, we develop a

continuous-time model for the underlying index and use it to obtain a semi-closed-form solution for the European option prices. In Section 3, we describe the data and the EMM methodology used to estimate the parameters of our continuous-time model. In Section 4, we bring the model to the data, estimate the risk premia implicit in 0DTE prices and illustrate the hedging implications of jumps. Section 5 concludes. The Appendices provide details of the derivations used throughout the paper.

2 Model

2.1 Dynamics for the underlying

The model is constructed to capture the salient features of intraday index dynamics and to accommodate popular option-pricing models as special cases. We will assume that the underlying index level S_t solves

$$\frac{\mathrm{d}S_t}{S_t} = \mu \mathrm{d}t + \sqrt{V_t}\ \mathrm{d}W_{1,t} + (e^{u_t} - 1)\,\mathrm{d}q_t - \lambda_t k \mathrm{d}t, \tag{1}$$

where the instantaneous variance V_t follows a mean-reverting diffusion given by the "square-root" specification of Heston (1993):

$$\mathrm{d}V_t = a\left(\bar{V} - V_t\right)\mathrm{d}t + \theta\sqrt{V_t}\ \mathrm{d}W_{2,t}. \tag{2}$$

In equations (1) and (2), μ is the expected return on the index, a is the speed of volatility's mean reversion, $\bar{V}$ is the long-run mean of the variance, and θ captures the volatility-of-volatility. The random vector $\mathbf{W}_t$ is a two-dimensional stochastic process consisting of correlated standard Brownian motions on the usual probability-space triplet $(\Omega, \mathcal{F}_t, \mathbb{P})$, where Ω is the event space, $\mathcal{F}_t$ is a Borel sigma-algebra on Ω and $\mathbb{P}$ is the "physical"

measure. The correlation coefficient between $W_{1,t}$ and $W_{2,t}$ is assumed constant and equal to ρ.

The term $(e^{u_t} - 1)\,\mathrm{d}q_t$ in equation (1) drives the jump component. The returns exhibit a jump at t if the jump flag $\mathrm{d}q_t$ equals one, which happens with probability $\lambda_t \mathrm{d}t$. We will assume that the jump intensity λ_t may change over time. In particular, we will allow the intensity to be a linear function of the instantaneous variance:

$$\lambda_t = \lambda_0 + \lambda_1 V_t. \tag{3}$$

Intuitively, if jumps arise more frequently in periods of high volatility, λ_1 should be positive. The significance of this term also implies that jumps appear in clusters, partially inheriting the behavior of V_t. The random variable u_t in equation (1) determines the relative magnitude of a jump.[1] The processes $\mathrm{d}q_t$ and u_t are independent, serially uncorrelated, and uncorrelated with the components of the Brownian motion vector $\mathrm{d}\mathbf{W}_t$. Also, neither $\mathrm{d}q_t$ nor u_t are $\mathcal{F}_t$-measurable.

It is reasonable to assume that the distribution of jump sizes is *not* concentrated around zero. However, this is not the case in most jump-diffusion specifications in the literature: jump sizes are usually modeled as random variables from a unimodal distribution. Jumps can be positive and negative and exhibit quite asymmetric behavior. Further, unimodal jump-size distributions imply that most jumps will be relatively small in magnitude, precisely the opposite of their nature. In such specifications, jumps are difficult to distinguish from continuously distributed returns generated by diffusion, leading to an overestimation of their frequency. We will therefore assume that jump sizes u_t are sampled from a mix-

[1] We apply the usual Bernoulli approximation for the Poisson counter $\mathrm{d}q_t$, allowing only zero or one jump per infinitesimal interval $\mathrm{d}t$, with respective probabilities of $1 - \lambda_t \mathrm{d}t$ and $\lambda_t \mathrm{d}t$. From the measuring point, several jumps within the same interval are equivalent to a single jump of cumulative magnitude captured by u_t.

ture of two normal distributions, one centered around a positive value, the other around a negative value:

$$u_t \;\sim\; p\,\mathcal{N}\left(\ln(1+k_+) - \sigma_+^2/2,\; \sigma_+^2\right) \;+\; (1-p)\,\mathcal{N}\left(\ln(1+k_-) - \sigma_-^2/2,\; \sigma_-^2\right). \qquad (4)$$

Hence, p is the probability of a positive jump, $k_+ \geq 0$ is the expected size of a positive jump, $k_- \leq 0$ is the expected size of a negative jump, while $\sigma_\pm^2 \geq 0$ is the variance of positive or negative jump size.

At time t, the expected contribution of jumps to return $\mathrm{d}S_t/S_t$ is

$$\mathbb{E}_t\left[(e^{u_t} - 1)\,\mathrm{d}q_t\right] = \lambda_t k \mathrm{d}t,$$

where

$$k \equiv pk_+ + (1-p)k_-.$$

Therefore, the return process is constructed such that jumps are, on average, compensated by the last term in equation (1).[2]

The outlined model specification has the form of a stochastic volatility jump-diffusion process with a bimodal distribution of jump sizes (hereinafter: SVJD-B).[3] It provides a robust and adaptable distributional framework. Skewed distributions may arise from non-zero average jumps or from correlations between return diffusion and volatility shocks. Likewise, fat-tailed distributions can result from stochastic volatility or a significant jump component. Moreover, the two explanations for skewness and kurtosis can be differentiated by the effects of the holding period. Stochastic volatility suggests a direct relationship

[2] We use $\mathbb{E}_t(\cdot)$ to denote the conditional expectation given the information available at time t, instead of a more cumbersome $\mathbb{E}(\cdot|\mathcal{F}_t)$.

[3] To the best of our knowledge, the bimodal assumption for the distribution of jump sizes was previously used only in a numerical valuation of real options in Dias and Rocha (2015).

between the holding period and the extent of excess skewness and kurtosis, whereas jumps indicate a strong inverse relationship.

Another convenient feature of the proposed model is that it contains several popular jump-diffusion and pure-diffusion benchmark models as its special cases. For example, by setting $p = 1$ and $\lambda_1 = 0$, we obtain the usual SVJD specification of Bates (1996, 2000). A stochastic volatility model without jumps (SV) of Taylor (2007) is obtained by setting all jump parameters (λ_0, λ_1, p, k_+, k_-, σ_+ and σ_-) to zero. Merton (1976) jump-diffusion model with constant variance is obtained by fixing $V_t = \bar{V}$, setting all stochastic-volatility parameters (a, θ, ρ) to zero, introducing a constant jump intensity ($\lambda_t = \lambda_0$, $\lambda_1 = 0$) and constraining the distribution of jump sizes to be unimodal ($p = 1$). Finally, the Black and Scholes (1973) model (BS) is obtained by additionally setting all jump parameters to zero.

Our SVJD-B model is specifically designed to capture the complex intraday dynamics relevant for 0DTE options while maintaining the analytical tractability necessary for efficient empirical implementation. We address several key structural choices and their underlying rationale.

The first is the absence of volatility jumps. Our variance process, based on the Heston (1993) framework, is modeled as a continuous diffusion (equation 2), implying that volatility changes smoothly, without discontinuities. While the literature widely acknowledges that price jumps can be accompanied by simultaneous jumps in volatility (co-jumps, as in Duffie, Pan, & Singleton, 2000; Eraker et al., 2003), our empirical analysis in Section 4 utilizes the swap-variance test of Jiang and Oomen (2008), which indicates that jumps in squared returns as a proxy for instantaneous variance are not statistically significant in our 5-minute SPX data. This empirical finding supports our modeling choice by suggesting that, for our high-frequency observations, price jumps constitute the dominant, sufficiently rich discontinuous component.

The second is the stationarity of structural parameters. The core structural parameters of our SVJD-B model, such as the mean-reversion speed a, the long-run variance $\bar{V}$ and the volatility-of-volatility θ, are treated as constant over the sample period. While 0DTE options are known to exhibit distinct intraday regimes, our auxiliary model, detailed in Section 3, explicitly filters out predictable diurnal volatility patterns before structural estimation. This deseasonalization ensures that the SVJD-B model focuses on capturing the underlying, non-diurnal risk drivers. Consequently, parameter stability in the structural model, combined with the state-dependent jump intensity λ_t in equation (3), provides a robust framework to model the unpredictable components of intraday dynamics without necessitating computationally burdensome time-varying structural parameters.

While more recent research, particularly on rough volatility (for example, Gatheral, Jaisson, & Rosenbaum, 2018), suggests that intraday volatility might exhibit "rougher" characteristics, our choice is driven by the need to maintain the model's crucial affine structure. This affine property is indispensable for deriving semi-closed-form solutions via Fourier inversion (Appendix B), which is essential for computationally efficient and large-scale calibration to the extensive 0DTE option dataset. Implementing rough volatility models, conversely, typically necessitates computationally intensive Monte Carlo simulations for option pricing, making empirical calibration across tens of thousands of option contracts highly challenging.

Finally, our model specifies a fixed bimodal distribution for jump sizes, equation (4). While a state-dependent jump size distribution, where jump magnitudes or tails could vary with instantaneous volatility, might offer additional realism (e.g., fatter tails in panic scenarios), such a specification would generally break the affine property of our model. The bimodal jump distribution, which distinguishes between positive and negative average jump sizes and their respective variances, already provides significant richness beyond standard

unimodal jump processes. This approach allows us to effectively capture the observed asymmetric tail behavior in 0DTE options, which is a primary focus of our study.

In essence, our SVJD-B model represents a strategic compromise: it balances sufficient realism to accurately capture the first-order effects of intraday jumps and tail risk in 0DTEs with the analytical tractability required for robust and efficient empirical calibration. Alternative modeling approaches, notably time-changed Lévy models, have also demonstrated considerable success. They effectively capture price jumps, stochastic volatility and volatility-return correlations (Carr & Wu, 2004), and can extract forward-looking correlation term structures from S&P and VIX derivatives (Ballotta, Eberlein, & Rayée, 2025). Nevertheless, we adopt the SVJD-B model for 0DTE option pricing, as it offers distinct advantages: it captures richer intraday volatility dynamics, separates diffusion and variance risks from jump risk, provides a more flexible fit to intraday skew and smile patterns, preserves a transparent economic interpretation of volatility versus jump effects and allows for more granular calibration to the short-maturity option surface.

2.2 0DTE option prices

A suitable property of the model introduced in Subsection 2.1 is that it can render a semi-closed-form solution for the price of European-style options with any maturity. To achieve this goal, we apply the usual risk-neutral pricing approach. By changing the probability measure, the risk-neutral counterparts of the processes for the return and the instantaneous variance from equations (1) and (2) become

$$\frac{\mathrm{d}S_t}{S_t} = (r_t - \delta_t)\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W^*_{1,t} + (e^{u_t} - 1)\,\mathrm{d}q_t - \lambda^*_t k^* \mathrm{d}t, \tag{5}$$

$$\mathrm{d}V_t = a^*\left(\bar{V}^* - V_t\right)\mathrm{d}t + \theta\sqrt{V_t}\,\mathrm{d}W^*_{2,t}, \tag{6}$$

where r_t is the instantaneous risk-free rate, and δ_t is the continuously compounded dividend yield.[4] The random vector $\mathbf{W}_t^*$ is now a standard Brownian motion under the risk-neutral probability measure $\mathbb{P}^*$. Its components have the same correlation coefficient ρ as under the physical measure $\mathbb{P}$. The mean-reversion speed of the instantaneous variance a^* and the expected jump size $\lambda_t^* k^* \mathrm{d}t$ depend on the market prices of volatility and jump risk, respectively. The explicit relationships are derived in Appendix A. The instantaneous risk premia are:

$$\begin{aligned} \text{return diffusion risk premium} &= \mu - (\bar{r} - \bar{\delta}), \\ \text{volatility risk premium} &= -(a - a^*)\bar{V}, \\ \text{overall jump risk premium} &= (\lambda_0 + \lambda_1 \bar{V})k - (\lambda_0^* + \lambda_1^* \bar{V}^*)k^*, \end{aligned} \tag{7}$$

where $\bar{r}$ is the mean risk-free rate and $\bar{\delta}$ is the mean dividend yield. The overall jump risk premium comprises the combined premia for uncertainty about a jump's arrival and size.

At the time t, the price of a European-style call option with the value of the underlying index equal to S_t, residual maturity τ and strike price X, is given by

$$C_t(S_t, V_t, \tau, X; \widehat{\psi}) = e^{-\delta_t \tau} S_t \Pi_1(S_t, V_t, \tau, X; \widehat{\psi}) - e^{-r_t \tau} X \Pi_2(S_t, V_t, \tau, X; \widehat{\psi}). \tag{8}$$

Similarly, the price of the corresponding put option is given by

$$P_t(S_t, V_t, \tau, X; \widehat{\psi}) = e^{-r_t \tau} X \left[1 - \Pi_2(S_t, V_t, \tau, X; \widehat{\psi})\right] - e^{-\delta_t \tau} S_t \left[1 - \Pi_1(S_t, V_t, \tau, X; \widehat{\psi})\right]. \tag{9}$$

The semi-closed-form expressions for the functions Π_1 and Π_2 are in Appendix B.

[4] For an index such as SPX, annual dividend yield represents total dividends that constituents pay in a year divided by the index value. Here, δ_t is the equivalent rate if the dividends were, hypothetically, paid continuously.

It is worth noting that equations (8) or (9) are not constrained to price only ultra-short-tenor options. They are, in fact, universal and can be applied to an arbitrary option maturity. However, the underlying index dynamics, as given by equations (1) and (2), emphasize the intraday features, making our model more suitable for shorter option maturities where these attributes have a substantial impact.

3 Data and methodology

3.1 Data

The method exploits intraday 0DTE price data information and the underlying index levels. For the 0DTEs, we employ five-minute bars from Cboe, focusing on the SPX Weeklys with root SPXW. These options are European and cash-settled at the close of the SPX index at 16:00 ET. The SPXW 0DTE data include the national best bid and offer (NBBO), with open/high/low/close (OHLC) prices, trade volumes, bid and ask sizes for the option, bid and ask prices of the underlying instrument, and implied volatility, as calculated by Cboe. We use all calls and puts traded on the following dates: January 31 (Wednesday), March 7 (Thursday), April 12 (Friday), May 13 (Monday), and June 18 (Tuesday), 2024, between 9:30 and 16:00 ET. The selection of these specific dates was carefully made to capture a comprehensive spectrum of intraday dynamics, intentionally including both days featuring high-impact macroeconomic events (such as the FOMC meeting on January 31) and more typical, non-event trading sessions. It also includes each weekday to avoid potential day-of-week biases. This diverse selection enables robust parameter estimation across varying levels of information flow and market uncertainty.

We begin with a dataset of 183,128 observations provided by Cboe for these dates. We remove rows with zero trading volume, zero bid or ask sizes and zero implied volatility,

leaving 22,410 observations in the final dataset. This final sample has 522 unique call contracts with 10,776 bars and 291 unique put contracts with 11,634 bars.

Table 1 shows the descriptive statistics of this dataset. The five-minute option volumes are relatively low, with an apparent left skew. There are occasional bursts in volume, with several instances of four-digit numbers of traded contracts. Over 80 percent of traded 0DTEs are in the strike range of ±2% of the active underlying price. The option prices span a wide range, from 0.05 to hundreds or even thousands of points. Most prices are, naturally, in the lower segment of this range due to ultra-short maturities and small relative moneyness. The residual maturities indicate that the sample spans the entire trading day, with time stamps between 9:35 and 16:00 ET. The median residual maturity is 3 hours 35 minutes for the calls and 3 hours 40 minutes for the puts.

Table 1: SPXW descriptive statistics. The table shows descriptive statistics for SPXW 0DTE five-minute data from Cboe. We use 22,410 observations of calls and puts traded on January 31, March 7, April 12, May 13 and June 18, 2024, between 9:30 and 16:00 ET. The rows show the trading volume, relative moneyness, the option's closing price and residual maturity. The prices are quoted in index points. The relative moneyness is the relative difference between the underlying and the strike price, $(S_t - X)/S_t$. The residual maturity is in hours. The columns represent the minimum, 10th percentile, median, 90th percentile, and maximum values.

	min	p10	p50	p90	max
Calls (10,776 observations)					
Volume	1.00	2.00	86.00	854.00	7154.00
Relative moneyness	−0.45	−0.02	0.00	0.01	0.03
Closing price	0.05	0.10	8.00	82.52	1634.60
Residual maturity (hrs)	0.00	0.92	3.58	6.00	6.42
Puts (11,634 observations)					
Volume	1.00	3.00	96.00	770.70	8068.00
Relative moneyness	−0.04	−0.02	0.00	0.01	0.24
Closing price	0.05	0.10	2.45	49.98	1522.50
Residual maturity (hrs)	0.00	0.92	3.67	5.92	6.42

For the underlying SPX index levels, we use the five-minute closing prices from Cboe's OHLC data for the period from January 2 to June 18, 2024. The selection of this period was motivated by its dynamic and event-rich market environment, which provides an ideal laboratory for studying intraday jumps. This half-year was characterized by significant shifts in monetary policy expectations, moving from anticipated early interest rate cuts to a "higher for longer" narrative from the Federal Reserve due to persistent inflation and a resilient US labor market. Alongside regular FOMC meetings that spurred market uncertainty, the period saw numerous high-impact macroeconomic data releases and major earnings announcements, particularly from AI-driven technology stocks. These conditions collectively generated frequent and impactful intraday price movements, making the dataset highly suitable for analyzing the pricing and hedging implications of extreme, discontinuous price changes relevant to 0DTE options.

There are 79 bars in each of the 116 trading days, resulting in 9,164 SPX closing-price observations and $78 \times 116 = 9,048$ return observations. Table 2 shows descriptive statistics for the SPX returns and squared returns. The average five-minute log return is approximately 0.06 basis points (bp), which is slightly below 5 bp when scaled to a daily level. The corresponding return volatility is 6.10 bp (approximately 54 bp per day when scaled), translating to an average squared return of 37.2. The intraday return distribution is left-skewed and leptokurtic. The squared returns are thus heavily right-skewed and fat-tailed. The skewness and kurtosis are also apparent from the distribution percentiles: the 10th percentile is higher in absolute value than the 90th percentile. However, the 90th-squared-return percentile is almost 10 times the median.

The rows labeled LB(L) show the values of the Ljung-Box statistics, capturing the autocorrelation of (squared) returns for lags $L = 1$, 6, and 12 bars (i.e., 5 minutes, 30 minutes, and 1 hour). The null hypothesis of no autocorrelation in returns cannot be rejected

Table 2: SPX descriptive statistics. The table shows descriptive statistics of SPX returns and squared returns. We use 9,048 log returns calculated from the five-minute SPX closing prices between January 2 and June 18, 2024, expressed in basis points. The first four rows show statistical moments. The next three rows indicate the 10th, 50th and 90th percentiles of the distribution. The rows labeled as LB(L) show the values of the Ljung-Box statistics for lags $L = 1$, 6 and 12. The rows labeled ADF(L) show the values of the Augmented Dickey-Fuller statistics for $L = 0$, 1, 6 and 12 lagged changes in (squared) returns. The last row displays values of the swap-variance test statistics of Jiang and Oomen (2008). The asterisks indicate the usual significance levels: *** for significance at 1%; ** for significance at 5%; * for significance at 10%.

	Returns (y_t)	Sq. returns (y_t^2)
Mean	0.06	37.17
St. dev.	6.10	97.16
Skewness	−0.37	9.10
Kurtosis	7.85	140.00
p10	−6.73	0.31
p50	0.21	9.25
p90	6.62	88.17
LB(1)	0.08	321.20***
LB(6)	11.92*	1042.10***
LB(12)	31.28***	1636.94***
ADF(0)	−95.42***	−70.44***
ADF(1)	−66.82***	−48.22***
ADF(6)	−35.17***	−21.04***
ADF(12)	−24.50***	−13.51***
SwV	−1.37*	0.00

for $L = 1$, while for $L = 6$, it can only be rejected at a 0.10 significance level. On the other hand, autocorrelation in squared returns is highly significant for all lags, indicating the usual presence of strong conditional heteroskedasticity. The values of the Augmented Dickey-Fuller (ADF) statistics indicate that the unit root hypothesis is convincingly rejected in favor of stationary returns and squared returns for $L = 0$, 1, 6 and 12 lagged changes in the observed series. Stationarity is a prerequisite for any approach based on the

method of moments, which critically relies on the stability of the data-generating process.

The last row displays values of the swap-variance test statistics of Jiang and Oomen (2008) for detecting jumps in returns and squared returns. The swap-variance test exploits the impact of jumps on the third and higher-order moments of asset returns. The test is based on the following statistic

$$\frac{T}{\sqrt{\mathscr{V}}}\left(SwV_T - RV_T\right) \xrightarrow{d} \mathcal{N}(0,1),$$

where

$$SwV_T = 2\sum_{t=2}^{T}\left(R_t - y_t\right)$$

is twice the accumulated difference between discretely and continuously compounded returns, $R_t = S_t/S_{t-1} - 1$ and $y_t = \ln(S_t/S_{t-1})$, respectively,

$$RV_T = \sum_{t=2}^{T} y_t^2$$

is the realized variance, and $\mathscr{V}$ is the asymptotic variance of the test statistic. A robust estimator of $\mathscr{V}$ can be calculated as

$$\widehat{\mathscr{V}} = \frac{1}{9}\frac{\mu_6}{\mu_1^6}\frac{T^3}{T-5}\sum_{t=0}^{T-6}\prod_{s=1}^{6}|y_{t+s}|,$$

with $\mu_n = \mathbb{E}(|y|^n)$. The null hypothesis of the swap-variance test is that S_t follows a process without jumps. Intuitively, the test statistic reflects the cumulative gain from a variance swap replication strategy, which is minimal in the absence of jumps but substantial in their presence. If the underlying process is continuous, the difference between SwV_T and RV_T approaches zero asymptotically. The results of this test indicate that jumps in returns are significant at the 0.10 level, while the jumps in squared returns are not. They justify the

choice of a simpler structural model, assuming that jumps occur only in the underlying returns (equation 1) rather than in instantaneous variances (equation 2). The results for the auxiliary model in Section 4 will additionally emphasize this point.

The remaining data required for option pricing consists of the instantaneous risk-free rate r_t and the continuously compounded dividend yield δ_t. We calculate the instantaneous risk-free rate from the prices of 4-week US Treasury Bills, available from the US Department of the Treasury at a daily frequency. We apply a single rate for each respective day. To compute the dividend yield, we use monthly S&P 500 dividend data from the Shiller Data and adjust for continuous compounding, treating each month as a single dividend.

3.2 Estimation methodology

Estimation of a continuous-time model, such as one given by equations (1)–(2), is never straightforward when we bring it to discretely sampled data. The primary challenge is that closed-form expressions for a discrete transition density are rarely available. The presence of unobservable state variables, such as stochastic volatility, makes this task even more arduous. In principle, some form of maximum likelihood estimation might be feasible (see, for example, Lo, 1988), but it relies on computationally intensive numerical procedures that integrate out latent variables from the likelihood function. The problem becomes even more complicated when jumps are introduced into the model.

Several alternatives to the maximum likelihood method have been proposed to address computational inefficiency. Examples of simulation-based inference for jump-diffusion models can be found in Andersen, Chung, and Sørensen (1999), Andersen et al. (2002) and Duffie et al. (2000). Simulation approaches based on the method of moments are practical whenever it is possible to alleviate the problem of inefficient inference, which can be achieved by carefully selecting moment conditions. For example, Pan (2002) employs the

simulated moments estimator (SME) of Duffie and Singleton (1993) to match sample moments with simulated ones, thereby estimating risk premia embedded in options on a stock market index. The efficient method of moments (EMM) of Gallant and Tauchen (1996) refines the SME approach by a convenient choice of moment conditions: they are obtained from the expected score of the auxiliary model. The auxiliary model is a discrete-time approximation of the sample distribution. Hence, there are at least two attractive features of the EMM approach: first, it will achieve the efficiency of the maximum likelihood technique under reasonable assumptions, and second, the objective function can be used to test for overidentifying restrictions, as with the ordinary generalized method of moments.

We use the EMM to estimate the proposed continuous-time model given by equations (1) and (2) and compare the alternatives. As pointed out by Andersen et al. (2002), the EMM procedure critically relies on the correct specification of the auxiliary model. The auxiliary model should approximate the conditional distribution of the return process as closely as possible. If the auxiliary model's score asymptotically spans that of the actual model, the EMM is asymptotically efficient (see Gallant & Long, 1997, for the proof). Therefore, any auxiliary model should capture the dominant features of the return dynamics in a discrete-time series. Specifically, it should be able to account for autocorrelation and heteroskedasticity, and model any excess skewness and kurtosis. Here, we apply a semi-nonparametric (SNP) specification for the auxiliary model by Gallant and Nychka (1987), based on the notion that a polynomial expansion can capture higher-order distribution moments.

Auxiliary model

For the auxiliary model, we use the diurnal adjustment from the Autoregressive Conditional Duration (ACD) model of Engle and Russell (1998) to capture deterministic diurnal

patterns in intraday index data. By allowing jumps to occur more frequently during periods of high volatility, we can effectively gauge their prominence as sensitive to diurnality. In particular, we expect the jumps to occur more often at the beginning of trading, consistent with the U-shaped intraday volatility pattern (Christensen, Hounyo, & Podolskij, 2018; Harris, 1986; Wood, McInish, & Ord, 1985) and evidence that around 70% of jumps in the S&P 500 occur during the first 30 minutes of trading (Bollerslev, Law, & Tauchen, 2008; Lee, 2011).

We start from the de-meaned log returns $y_t \equiv \ln(S_t/S_{t-1})$, i.e.,

$$\varepsilon_t = y_t - \bar{y}.$$

These return innovations follow:

$$\begin{aligned} \varepsilon_t &= s_t \eta_t \\ \eta_t &= z_t \sqrt{h_t}, \quad z_t \sim \text{i.i.d. } \mathcal{N}(0,1). \end{aligned}$$

Here, s_t is a deterministic seasonal component in intraday log volatility, and h_t is the conditional variance. We assume that the log of the squared seasonal component has a simple quadratic polynomial form:

$$\ln s_t^2 = b_0 + b_1 H_t + b_2 H_t^2, \tag{10}$$

where H_t are daily hours until time t, while the conditional variance of deseasonalized residuals η_t follows a GJR(L_1,L_2) process of Glosten, Jagannathan, and Runkle (1993):

$$h_t = \omega + \sum_{l=1}^{L_1} \alpha_l \eta_{t-l}^2 + \sum_{l=1}^{L_2} \beta_l h_{t-l} + \sum_{l=1}^{L_1} \gamma_l \mathbf{1}_{\{\eta_{t-l}<0\}} \eta_{t-l}^2. \tag{11}$$

The terms associated with parameters α_l, β_l and γ_l are the usual ARCH, GARCH and leverage terms, respectively, and **1** is the indicator function.

Finally, to accommodate the presence of fat tails in the intraday return distribution remaining after removing the diurnal component and conditional heteroskedasticity, we follow Gallant and Nychka (1987) and Gallant and Tauchen (1996) and augment the conditional probability density function of the auxiliary model by a polynomial in standardized returns:

$$P_K(z_t) = \sum_{i=0}^{K} c_i z_t^i, \tag{12}$$

with $c_0 = 1$.[5] Thus, the vector of auxiliary model parameters

$$\varphi = [\bar{y}\ b_0\ b_1\ b_2\ \omega\ \alpha_1\ \ldots\ \alpha_{L_1}\ \beta_1\ \ldots\ \beta_{L_2}\ \gamma_1\ \ldots\ \gamma_{L_1}\ c_1\ c_2 \ldots\ c_K]$$

has $n_\varphi = 5+2L_1+L_2+K$ components consisting of the mean return, the ACD parameters, the GJR parameters and the SNP polynomial parameters. We estimate φ via a maximum likelihood estimator. The log-likelihood function has the following form:

$$\ln \mathcal{L}\left(\varphi | y_1, y_2, \ldots, y_T\right) = \sum_{t=2}^{T} \ln f(y_t | \mathcal{F}_{t-1}; \varphi), \tag{13}$$

where T is the number of index return observations and

$$f(y_t | \mathcal{F}_{t-1}; \varphi) = \frac{P_K^2(z_t)}{\int_{-\infty}^{\infty} P_K^2(z) e^{-z^2/2} \mathrm{d}z} \frac{e^{-z_t^2/2}}{s_t \sqrt{h_t}} \tag{14}$$

is the conditional density function.

[5] In principle, one can extend the polynomial in equation (12) to depend on the lagged standardized residuals, as suggested by Andersen et al. (1999). However, as in Andersen et al. (2002), these additional terms were insignificant when applied to our data and would lead to overparameterization while introducing an unnecessary layer of complexity to the model.

EMM: numerical simulation and model parameters

The EMM estimation step operates as follows. Given the vector of structural parameters

$$\psi = \left[\mu \; a \; \bar{V} \; \theta \; \rho \; \lambda_0 \; \lambda_1 \; p \; k_+ \; k_- \; \sigma_+ \; \sigma_-\right],$$

we simulate the sample of index levels $\{\widetilde{S}_t\}_{t=1}^{T_{\text{sim}}}$ and instantaneous variances $\{\widetilde{V}_t\}_{t=1}^{T_{\text{sim}}}$ using the specification given by the continuous-time model, equations (1) and (2). The EMM estimator of structural parameters ψ is defined as

$$\widehat{\psi} = \arg\min_{\psi} \; \mathbf{m}(\psi, \widehat{\varphi})' \; \widehat{\mathbf{I}}^{-1} \; \mathbf{m}(\psi, \widehat{\varphi}), \tag{15}$$

where

$$\mathbf{m}(\psi, \widehat{\varphi}) = \frac{1}{T_{\text{sim}}} \sum_{t=1}^{T_{\text{sim}}} \frac{\partial \ln f(\widetilde{y}_t | \mathcal{F}_{t-1}; \widehat{\varphi})}{\partial \varphi} \tag{16}$$

is the expectation of the score function evaluated as the sample mean across simulated returns $\widetilde{y}_t \equiv \ln(\widetilde{S}_t / \widetilde{S}_{t-1})$, $\widehat{\varphi}$ are estimated auxiliary model parameters, and

$$\widehat{\mathbf{I}} = \frac{1}{T_{\text{sim}}} \sum_{t=1}^{T_{\text{sim}}} \frac{\partial \ln f(y_t | \mathcal{F}_{t-1}; \widehat{\varphi})}{\partial \varphi} \frac{\partial \ln f(y_t | \mathcal{F}_{t-1}; \widehat{\varphi})'}{\partial \varphi} \tag{17}$$

is a consistent estimator of the asymptotic covariance matrix, $\mathbf{I}$, of the sample score vector. As in Andersen et al. (1999), we estimate $\widehat{\mathbf{I}}$ directly from the auxiliary score using the observed data y_t to avoid the need for computation of the weighting matrix using the simulated data $\widetilde{y}_t$ in the EMM estimation step. The matrix $\widehat{\mathbf{I}}$ is an $n_\varphi \times n_\varphi$ square matrix.

The EMM test of a correct model specification follows directly from equation (15). The test statistic is T_{sim} times the minimized objective function, distributed as χ^2 with $n_\varphi - n_\psi$ degrees of freedom, where n_ψ is the number of components of ψ. Under the null hypothesis, the model is correctly specified. We utilize the complete analytical form of the

auxiliary log-likelihood gradient to accelerate convergence in equation (15). The closed-form expressions for its components are given in Appendix C. If the test of overidentifying restrictions is rejected, these components may indicate directions in which the structural model fails to accommodate the data.

In the simulation step, we sample at time intervals of 1/300 of a bar, i.e., once every second. Combined with the Euler discretization of equations (1) and (2), this approach allows for a sufficiently smooth sample path. We create a sample of 120 days at each run, each with 79 five-minute bars and 300 subintervals. To eliminate the effects of initial conditions, particularly for stochastic volatility V_t, we discard the "burn-in" period of the first 20 days. The final simulated sample, $\{\widetilde{y}_t\}_{t=1}^{T_{\text{sim}}}$, of $T_{\text{sim}} = 7,900$ five-minute observations was calculated from the "closing" prices $\widetilde{S}_t$ at the end of each simulated bar.[6]

EMM: standard errors

The EMM standard errors are obtained as the square roots of the diagonal elements of the asymptotic covariance matrix. The consistent estimator of this matrix is

$$\text{cov}(\widehat{\psi}) = \frac{1}{T_{\text{sim}}} \left[\frac{\partial \mathbf{m}(\widehat{\psi}, \widehat{\varphi})'}{\partial \psi} \widehat{\mathbf{I}}^{-1} \frac{\partial \mathbf{m}(\widehat{\psi}, \widehat{\varphi})}{\partial \psi} \right]^{-1}. \tag{18}$$

We numerically calculate the vector of partial derivatives with respect to the structural parameters ψ from the analytical score using the finite-difference method. These partial derivatives form an $n_\varphi \times n_\psi$ Jacobian matrix of the log-likelihood. To improve the numerical stability, we perform a principal component analysis of the $n_\psi \times n_\psi$ matrix inside the square bracket in equation (18) before computing its inverse. We eliminate the components with eigenvalues below 10^{-4} and retain the ones corresponding to the $n \leq n_\psi$ highest eigenvalues.

[6] The simulation produces $120 \times 79 \times 300 = 2,844,000$ observations. After we discard the first 20 days, the sample consists of $100 \times 79 \times 300$ observations. We then choose every 300th subinterval to sample a densely simulated path at five-minute intervals. This procedure leaves 100×79 simulated observations.

We decompose the $n_\psi \times n_\psi$ matrix using the spectral theorem to a product of the form UDU', where U is an $n_\psi \times n$ reduced unitary matrix and D is an $n \times n$ reduced diagonal matrix of non-negligible eigenvalues. This product is significantly easier to invert, resulting in numerically stable elements of the asymptotic covariance matrix and plausible, realistic standard errors.

Risk-neutral parameters

The risk-neutral parameter estimates are required to match the model-implied option prices with the observed ones. We consider the risk-free rate r_t and the dividend yield δ_t exogenous (see subsection 3.1). The risk-neutral parameters, a^*, λ_0^* and k^* can be obtained by solving

$$\min_{\{a^*,\lambda_0^*,k^*\}} \sum_{i\in\mathscr{S}_{\text{call}}} \frac{(C_i^{\text{model}} - C_i^{\text{data}})^2}{C_i^{\text{data}}} + \sum_{i\in\mathscr{S}_{\text{put}}} \frac{(P_i^{\text{model}} - P_i^{\text{data}})^2}{P_i^{\text{data}}}, \tag{19}$$

where $\mathscr{S}_{\text{call}}$ and $\mathscr{S}_{\text{put}}$ denote the call and put subsets from the final dataset. The remaining risk-neutral parameters $\bar{V}^*$ and λ_1^* that affect option prices and implied volatilities follow trivially from the constraints between the other risk-neutral parameters and their physical counterparts, as demonstrated in Appendix A.

The estimator given by equation (19) is designed to minimize the squared difference between the prices obtained from the data and the semi-closed-form option prices from equations (8) and (9). For every contract i, the mid-points between bid and ask prices at the end of five-minute intervals are used as C_i^{data} and P_i^{data}. The objective function in equation (19) is distributed as χ^2 with $\#\mathscr{S}_{\text{call}} + \#\mathscr{S}_{\text{put}} - 1$ degrees of freedom.

Estimation steps visualized

Figure 2 visualizes the steps in the estimation procedure. The calibration of jump parameters is embedded within the EMM step. Specifically, the diurnal coefficients b_0, b_1, and b_2,

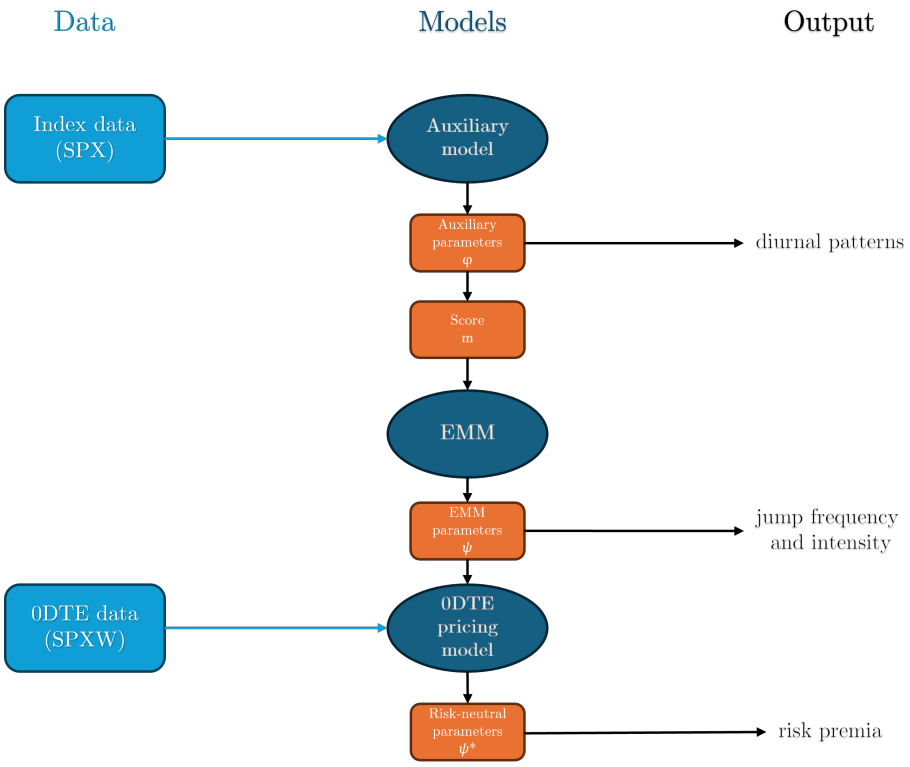


Figure 2: Steps in the estimation procedure

estimated from high-frequency index data, capture the systematic intraday volatility variation. These coefficients, along with the auxiliary model parameters φ, are used to construct the analytical gradient of the likelihood function, which forms the expected score vector $\mathbf{m}$. The structural parameters ψ, which include jump intensity and size, are then estimated by minimizing the weighted sum of squared components of $\mathbf{m}$, using simulated data based on the discretized versions of stochastic processes defined in equations (1) and (2). The 0DTE option data are used in equation (19) to infer the risk-neutral model parameters. Finally, these parameters are combined with $\widehat{\psi}$ to infer the risk premia.

This setup enables the model to flexibly adjust jump parameters throughout the trading

day, with intensity levels sensitive to intraday volatility, regardless of its diurnal structure. As a result, the calibration does not rely on manually specified time blocks but rather reflects the empirical time variation inherent in the index data.

4 Results

4.1 Estimation of the auxiliary model

Table 3 reports the results of the SNP step. It shows the maximum likelihood parameter estimates $\widehat{\varphi}$ and their standard errors. We run the estimations over all possible combinations of the triplet (L_1, L_2, K), with each value ranging from 0 to 10. Selection based on the Bayesian information criterion indicates that the best-fitting auxiliary model is the one in which $L_1 = 1$, $L_2 = 2$ and $K = 6$, making the total number of auxiliary-model parameters equal to $n_\varphi = 15$. The last column of Table 3 shows the scores under the optimal parameter values $\widehat{\varphi}$ and their sample standard errors. It is easy to verify that none of the scores has a significant t statistic.

Table 4 shows descriptive statistics for the standardized and squared standardized residuals after applying the optimal auxiliary model with parameters from Table 3. The moments and the percentiles indicate that the auxiliary model reduces most skewness and kurtosis in returns. The Ljung-Box statistics confirm that the optimal model successfully removes the systematic first- and second-order dependencies. Stationarity is also preserved, as indicated by the values of the ADF statistics. Finally, the swap-variance test now indicates a strong presence of jumps in the underlying returns, while the instantaneous variances do not exhibit such behavior. The fact that jumps become more distinct after filtering out diurnal patterns and heteroskedasticity is crucial for assessing the role of jumps in risk premia.

Table 3: Estimation results for the auxiliary model. The table shows the maximum likelihood parameter estimates $\widehat{\varphi}$ of the auxiliary model described in subsection 3.2 and the corresponding scores. The parameters represent the mean return ($\bar{y}$), the ACD parameters (b_i), the GJR parameters (ω, α_1, β_1, β_2 and γ_1) and the SNP polynomial parameters (c_i). The numbers in parentheses are the standard errors.

Parameter	Coefficient	Score
$\bar{y}$	0.0912	−0.0010
	(0.1123)	(0.0025)
b_0	12.5990	0.0007
	(0.9810)	(0.0028)
b_1	−1.5987	0.0057
	(0.1515)	(0.0409)
b_2	0.0620	0.0836
	(0.0059)	(0.6395)
ω	0.0718	0.0122
	(0.0171)	(0.0391)
α_1	0.0825	−0.0154
	(0.0197)	(0.0088)
β_1	0.3233	0.0473
	(0.0713)	(0.0825)
β_2	0.5609	0.0444
	(0.0771)	(0.0818)
γ_1	0.0478	−0.0207
	(0.0383)	(0.0185)
c_1	0.0435	−0.0125
	(0.0184)	(0.0231)
c_2	−0.1014	0.0170
	(0.0151)	(0.0358)
c_3	−0.0141	−0.0128
	(0.0059)	(0.0923)
c_4	0.0011	0.2371
	(0.0035)	(0.2287)
c_5	0.0004	−0.2804
	(0.0006)	(0.6887)
c_6	0.0011	0.5578
	(0.0002)	(2.2011)

Table 4: Descriptive statistics for the standardized residuals. The table shows descriptive statistics for the standardized residuals (z_t) and squared standardized residuals (z_t^2) after applying the optimal auxiliary model with parameters shown in Table 3. The rows are the same as in Table 2: the first four display statistical moments, and the following three indicate the 10th, 50th, and 90th percentiles of the distribution. The rows labeled as LB(L) show the values of the Ljung-Box statistics for lags $L = 1$, 6 and 12. The rows labeled ADF(L) show the values of the Augmented Dickey-Fuller statistics for $L = 0$, 1, 6 and 12 lagged changes in (squared) returns. The last row displays values of the swap-variance test statistics of Jiang and Oomen (2008). The asterisks indicate the usual significance levels: *** for significance at 1%; ** for significance at 5%; * for significance at 10%.

text	Std. residuals (z_t)	Sq. std. residuals (z_t^2)
Mean	0.00	0.90
St. dev.	0.95	2.20
Skewness	−0.25	13.41
Kurtosis	6.99	370.14
p10	−1.12	0.01
p50	0.02	0.29
p90	1.08	2.19
LB(1)	0.06	0.74
LB(6)	2.56	4.49
LB(12)	14.75	10.56
ADF(0)	-95.37^{***}	-81.71^{***}
ADF(1)	-67.23^{***}	-55.16^{***}
ADF(6)	-35.58^{***}	-23.53^{***}
ADF(12)	-24.78^{***}	-14.69^{***}
SwV	97.00^{***}	−0.14

Figure 3 illustrates the U-shaped diurnal pattern in intraday SPX returns described by equation (10) under the optimal auxiliary model. The red line, corresponding to the $\ln s_t^2$ pattern, follows the usual parabolic shape and captures the general intraday seasonality quite well. Thus, the SPX intraday volatility peaks near the opening and closing hours. By controlling for this effect, we ensure that the intraday jumps are not artificially over-represented at the beginning and the end of each day. This way, the EMM estimates will

adequately attribute the jump frequency and size parameters to genuine return discontinuities, which are unpredictable, rather than excessive volatility during specific trading hours, which is predictable. The absence of jumps in volatility, indicated by the last row in Table 4, facilitates this attribution.

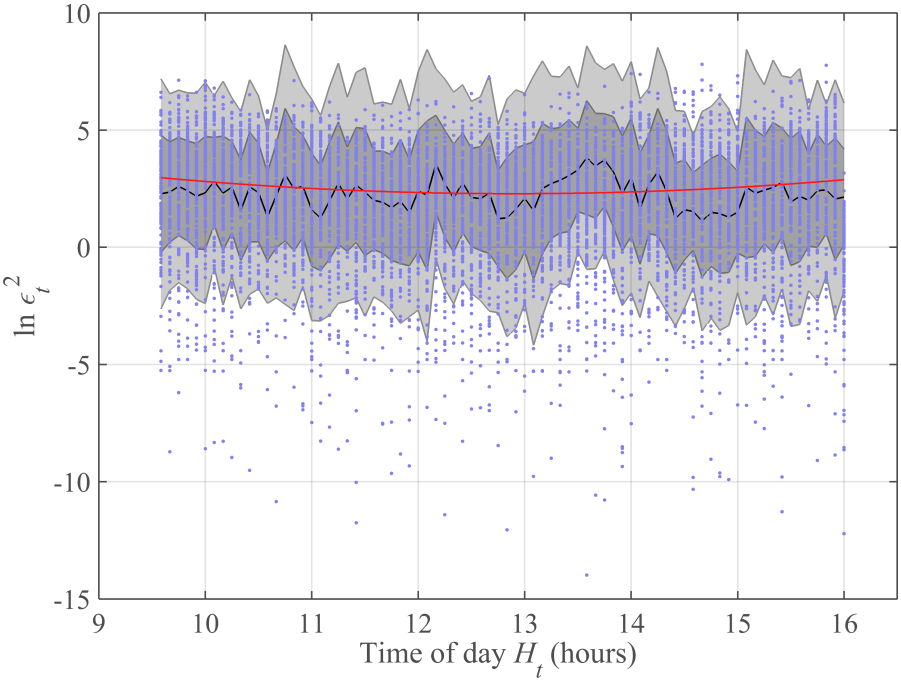


Figure 3: Diurnal pattern in intraday SPX returns. We use 9,048 log-returns y_t calculated from the five-minute SPX closing prices between January 2 and June 18, 2024. The figure shows the trading hour patterns of the log of squared return innovations $\ln \varepsilon_t^2$ (blue dots) as a function of the time of day H_t. The log of the squared seasonal component, $\ln s_t^2$ (red line), follows a quadratic polynomial form given by equation (10). The parameters $\bar{y}$ and b_i are from the optimal auxiliary model whose estimates are given in Table 3. The thick black line shows the median value of $\ln \varepsilon_t^2$. The shaded areas represent one and 1.96 standard deviations of $\ln \varepsilon_t^2$ around the median.

4.2 EMM estimation

With the optimal score parameters $\widehat{\varphi}$ obtained from the auxiliary model, we can run the EMM step and obtain the structural parameter estimates $\widehat{\psi}$ using equations (15)–(17). Table 5 summarizes the estimation results obtained from the SVJD-B and the alternative models.

We can draw several important conclusions from these estimates. As expected, stochastic volatility is important: the constant-volatility Merton (1976) and Black and Scholes (1973) models can be rejected as the p-values in the Hansen (1982) test of overidentifying restrictions are below 0.01. Jumps play a vital role in describing the SPX intraday dynamics – the Taylor (2007) SV model without jumps can be rejected at a 0.01 significance level, unlike the SVJD-B and SVJD specifications. The SVJD-B model with $\rho = 0$ is the only one that attains a p-value above 0.05, suggesting that the correlation between return and volatility innovations is perhaps not essential but that the quality of fit is undoubtedly improved by including a bimodal distribution of jump sizes and an affine jump intensity specification as in equation (3).

The intraday jumps are significant in frequency and magnitude. The values of λ_0 in SVJD-B models indicate an average of about 4.5 jumps per day. This frequency increases with volatility in the complete model and the specification with $\rho = 0$. The value of $\lambda_1 \approx 17.9$ implies that an increase of 10 bp in daily volatility corresponds to a 0.0014 increase in jump frequency. Since jump intensity is only slightly higher during high-volatility periods, jump clustering near market opening and closing hours may not entirely be a consequence of diurnal volatility patterns. Negative jumps are slightly more likely than positive ones, as $p < 0.5$ in all three SVJD-B specifications. In the full specification, positive jumps occur approximately 45 percent of the time. This asymmetry captures part of the skewness of the unconditional return distribution. The average and the standard deviation of jump sizes are

also greater in absolute terms for the negative than for the positive jumps. The magnitude of positive jumps is around 14.5 bp on average, with a standard deviation of 6.3 bp. The average of negative jumps is -15.9 bp, with a standard deviation of 7.1 bp. Therefore, on average, positive and negative jumps move the index by about 2.5 standard deviations in five-minute returns when they occur. Their distributions around these averages have characteristic widths comparable to the entire return distribution. Overall, the expected jump size is $k = -2.1$ bp.

To verify these findings independently, we run a nonparametric jump test of Lee and Mykland (2007). At a 0.05 significance level, the test detected 492 jumps, of which 211 were positive and 281 were negative. These numbers correspond to 4.24 jumps per day, with a 0.43 probability of encountering a positive jump. Conditionally on an intraday return observation being flagged as a jump by the test, its average value is -2.83 bp. Positive returns detected as jumps have an average value of 14.49 bp, while the negative ones have an average value of -15.83 bp.

Our jump frequency and magnitude may appear more frequent and smaller than estimates from some prior studies. The discrepancy primarily stems from the fundamental difference in data granularity. Our study focuses on *intraday* data and 0DTE options, whereas the earlier studies predominantly analyze daily or yearly jump occurrences. A single daily jump typically manifests as multiple, smaller, discontinuous price movements when observed at 5-minute frequency, which accumulate into a larger daily event.

Table 5: Estimation results for the EMM. The table shows the EMM estimates of the structural parameters ψ, obtained via procedure described in subsection 3.2. The simulated sample consists of $100 \times 79 \times 300$ observations after discarding the burn-in period of 20 days. We then choose every 300th subinterval to match the five-minute periods of the score function, resulting in $7,900$ simulated observations. The table columns represent the estimates for the full SVJD-B model, given by equations (1) and (2), and its various restrictions: jump intensity independent of instantaneous variance ($\lambda_1 = 0$), uncorrelated Brownian motions ($\rho = 0$), the Bates (1996) SVJD specification with a unimodal distribution of jumps, the Taylor (2007) stochastic volatility model without jumps (SV), the Merton (1976) jump-diffusion model and the Black and Scholes (1973) simple diffusion model (BS). The numbers in parentheses are the standard errors. The last two rows show the results of the Hansen (1982) test of overidentifying restrictions: chi-square test statistics with the degrees of freedom (DoF) in the bracket and the corresponding p-values.

Parameter	SVJD-B			SVJD	SV	Merton	BS
	Full model	$\lambda_1 = 0$	$\rho = 0$				
μ	0.0685	0.0757	0.0790	0.0772	0.1750	0.0967	0.1826
	(0.0125)	(0.0095)	(0.0243)	(0.0060)	(0.0523)	(0.0003)	(0.0381)
a	0.0543	0.0594	0.0419	0.0785	0.0170		
	(0.0372)	(0.0653)	(0.0474)	(0.0581)	(0.0030)		
$\bar{V}$	38.5838	38.5765	38.5900	38.5724	37.8745	38.6039	17.2671
	(0.0794)	(0.1459)	(0.3304)	(0.1295)	(0.0009)	(0.4164)	(0.6706)
θ	1.1136	0.7511	0.6862	1.1341	0.8079		
	(0.3955)	(0.3334)	(0.2649)	(0.4127)	(0.0659)		
ρ	0.1933	−0.0336		−0.0691	−0.0503		
	(0.1145)	(0.1434)		(0.0883)	(0.0599)		
λ_0	0.0573	0.0670	0.0638	0.0336		0.0461	
	(0.0261)	(0.0093)	(0.0302)	(0.0170)		(0.0041)	
λ_1	17.9390		17.9393				
	(0.0000)		(0.0000)				
p	0.4548	0.4935	0.4825				
	(0.0000)	(0.0193)	(0.0000)				
k_+	14.4667	14.3959	14.4609				
	(0.7226)	(0.3668)	(0.8758)				
k_-	−15.8950	−15.9670	−15.8981	−2.8494		−2.8663	
	(0.4428)	(0.5038)	(0.4742)	(0.1836)		(0.4663)	
σ_+	6.3167	6.3695	6.3263				
	(0.5683)	(0.3156)	(0.3895)				
σ_-	7.0731	7.0662	7.1182	16.4288		10.7458	
	(0.3455)	(0.3354)	(0.3953)	(0.6839)		(0.4179)	
χ^2 [DoF]	9.3646 [3]	13.0875 [4]	8.8477 [4]	14.7347 [7]	24.6288 [10]	23.5101 [10]	30.8530 [13]
p-value	0.0248	0.0109	0.0650	0.0396	0.0061	0.0090	0.0035

Our findings for average positive and negative jump sizes are therefore reasonably large in an intraday context. They are significant relative to the 54 bp daily return volatility and are precisely the kind of micro-events that drive 0DTE price changes. Indeed, studies explicitly focusing on intraday data, such as Lee (2011), reporting 27.87 jumps per year for SPY, albeit in highly volatile periods, and Bollerslev et al. (2008), documenting large jumps for individual stocks, provide evidence that price discontinuities are far more frequent at higher sampling frequencies. While Andersen et al. (2017) also discuss weekly options and aggregate negative jumps larger than -1.5% annually, they too acknowledge varying frequencies in different market regimes. Consequently, our estimates accurately reflect the true intraday jump process relevant to the pricing and hedging of ultra-short-dated derivatives, which are inherently exposed to high-frequency within-day jump events.

Using the structural parameter estimates from Table 5, we can infer the ex-post probability of a jump at any given moment implied by the actual data. Following Johannes (2004), we use the Gibbs sampling technique to compute the filtering distribution of jump times and sizes. The Gibbs sampler iteratively draws values from the filtering distribution of conditional variances, $\mathbb{P}(V_{t+\Delta t}|V_t, \Delta q_{t+\Delta t}, u_{t+\Delta t}, y_{t+\Delta t}, y_t; \widehat{\psi})$, the filtering distribution of jump times, $\mathbb{P}(\Delta q_{t+\Delta t}|u_{t+\Delta t}, y_{t+\Delta t}, y_t, V_{t+\Delta t}, V_t; \widehat{\psi})$, and the filtering distribution of jump sizes $\mathbb{P}(u_{t+\Delta t}|\Delta q_{t+\Delta t}, y_{t+\Delta t}, y_t, V_{t+\Delta t}, V_t; \widehat{\psi})$, all of which are know conditional distributions, where $\{y_t\}_{t=1}^T$ is the observed time series of daily returns and $\widehat{\psi}$ are the estimated SVJD-B parameters. In each iteration j, the algorithm produces a sequence $\{\{V_t^{(j)}\}_{t=1}^T, \{\Delta q_t^{(j)}\}_{t=1}^T, \{u_t^{(j)}\}_{t=1}^T\}$ of conditional variances, jump flags and jump sizes, which are drawn from the joint distribution, $\mathbb{P}(V_{t+\Delta t}, q_{t+\Delta t}, u_{t+\Delta t}|y_{t+\Delta t}, y_t, V_t; \widehat{\psi})$. We work with $N_{\text{sim}} = 200,000$ iteration steps and discard the "burn-in" period of the first 100,000 iterations.

Figure 4 illustrates the results. It shows the estimated jump sizes for SPX returns

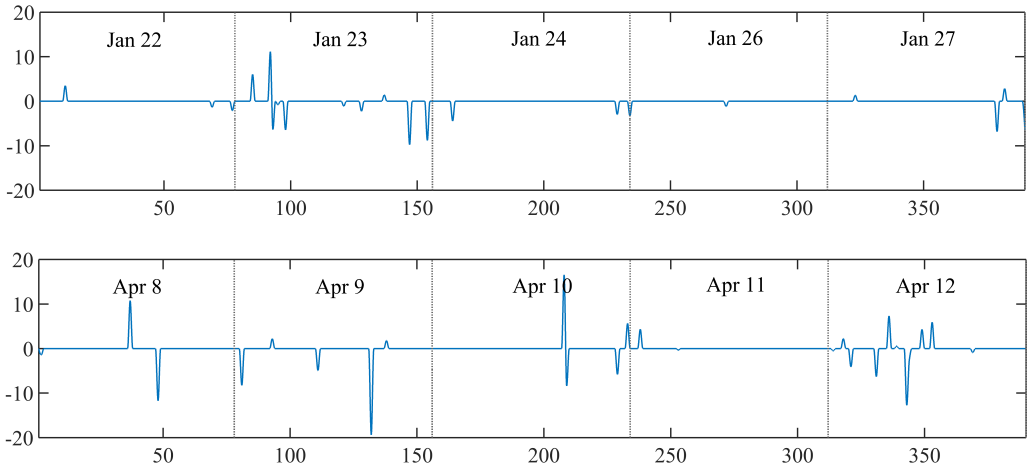


Figure 4: Estimated jump sizes. This figure shows the estimated jump sizes for intraday SPX returns during weeks 4 (top) and 16 (bottom) of 2024, i.e., January 22–26 and April 8–12, in basis points. The jump sizes are obtained via Gibbs sampling over 200,000 simulations, discarding the first 100,000 as a burn-in period. We iteratively sample from the filtering distributions of variances, jump times and jump sizes given the estimated structural parameters $\widehat{\psi}$ from Table 5 and the observed time series of intraday returns $\{y_t\}_{t=1}^{T}$. The estimated jump size is calculated as the average sample value of $\exp[u_{t+\Delta t}^{(j)} - 1]\Delta q_{t+\Delta t}^{(j)}$.

during January 22–26 and April 8–12, 2024. The estimated jump sizes are calculated as the average sample value of $\exp[u_{t+\Delta t}^{(j)} - 1]\Delta q_{t+\Delta t}^{(j)}$. We observe that jumps are *ex post* spread inhomogeneously throughout a typical week. There are days without a single jump and others with several noticeable discontinuities in SPX levels. The bimodal jump-size distribution in the SVJD-B model ensures that most identified jumps will be significant, a noteworthy feature of the model. In the standard SVJD specification with an unimodal distribution of jump sizes, most jumps would be challenging to identify, as they would be distributed around the relatively small unconditional expectation k.

The intraday distribution of jump sizes follows a more regular pattern. Figure 5 shows

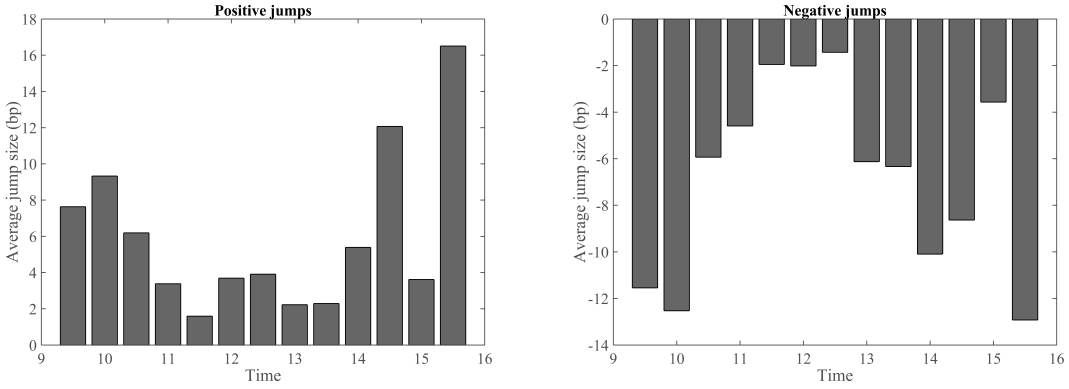


Figure 5: Distribution of estimated jump sizes. This figure shows the distribution of estimated positive (left) and negative (right) jump sizes for intraday SPX returns over the entire sample period in basis points. The bars show the average jump sizes across 30-minute periods during trading hours. The jump sizes are obtained via Gibbs sampling over 200,000 simulations, discarding the first 100,000 as a burn-in period. We iteratively sample from the filtering distributions of variances, jump times and jump sizes given the estimated structural parameters $\widehat{\psi}$ from Table 5 and the observed time series of intraday returns $\{y_t\}_{t=1}^{T}$. The estimated jump size is calculated as the average sample value of $\exp[u_{t+\Delta t}^{(j)} - 1]\Delta q_{t+\Delta t}^{(j)}$.

the estimated positive and negative jump size distribution over the entire sample period. The bars represent the average jump sizes obtained from the Gibbs sampler across 13 half-hour trading periods (9:30 to 16:00). The jumps are prominent, particularly at the beginning and end of trading, regardless of sign. For example, the average size of a positive jump is 8.5 bp within the first trading hour, 10.1 bp within the last trading hour and only 4.5 bp between the first and the last hour. Similarly, the average size of a negative jump is -12.0 bp within the first trading hour, -8.2 bp within the last trading hour and -5.2 bp in the meantime.

4.3 Risk premia implicit in SPXW prices

Intraday jumps lead to abrupt changes in SPX levels. Therefore, they should have a substantial impact on 0DTE option prices, as they can change their moneyness and profitability instantaneously. With a limited downside risk, investors should be willing to pay more to be exposed to an increased upside potential of substantial changes in the underlying index levels.

We can observe this impact directly through the implied volatility smiles. Figure 6 illustrates the annualized implied volatility smiles for option contracts having six and two hours until expiry as a function of relative moneyness $(S_t - X)/S_t$.[7] We compare the implied volatility smiles obtained from the SVJD-B model without the risk premia and the corresponding model with higher jump intensities and sizes. For the former case (blue lines), we assume the risk-neutral drift equals $r_t - \delta_t$, where the annualized instantaneous risk-free rate and the continuously compounded dividend yield are set to $r_t = 5.20\%$ and $\delta_t = 1.32\%$, respectively, matching the values from June 18, 2024, the last day of our sample. We also use the opening price on the same date for the underlying, $S_t = 5476.15$ and set the instantaneous variance equal to its long-term mean, $V_t = \bar{V}$. The remaining risk-neutral parameters are set to their full-model estimates under physical probability from Table 5.

We contrast this baseline case with the implied volatility obtained by increasing jump intensities and sizes by 20% while keeping all other parameters equal. This scenario, shown in red lines in Figure 6, confirms our intuition. Despite the ultra-short term of the contracts, the implied volatility increase is notable. The smile patterns are more pronounced at the shorter residual maturity of $\tau = 2$ hours, since a jump could be more impactful as the

[7] All curves were obtained by averaging the implied volatilities obtained from call and put prices with the same strike and maturity. The differences between call- and put-implied volatilities are negligible, of the order of 10^{-4} percent per annum.

remaining time to compensate for the shock diminishes. This behavior is the usual kurtosis effect on smiles, albeit more excessive. For both maturities, the effect of increased jump risk is slightly higher for $S_t < X$: negative jumps are more likely, more increased on average in absolute terms, and more widely spread, so this distributional asymmetry only becomes exaggerated.

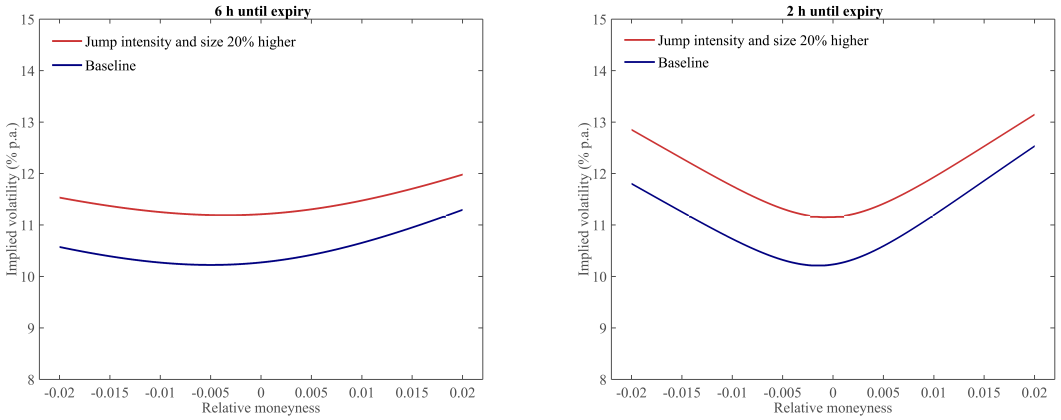


Figure 6: The impact of jumps on 0DTE implied volatility. This figure shows the annualized implied volatility smiles for 0DTE options with six (left) and two (right) hours until expiry. The blue lines represent the implied volatility smiles obtained from the SVJD-B model without the risk premia, i.e., by setting the drift parameter to $r_t - \delta_t = 3.9480\%$ and the remaining risk-neutral parameters equal to their full-model structural parameter estimates from Table 5. The red lines are obtained by assuming that jump intensities and sizes are each 20% higher. Both sets of curves were obtained by averaging the implied volatilities obtained from call and put prices with the same strike and maturity. The horizontal axis is the relative moneyness $(S_t - X)/S_t$.

We now apply our option-pricing model to the 0DTE data described in subsection 3.1, consisting of 22,410 observations of SPXW calls and puts traded on January 31, March 7, April 12, May 13 and June 18, 2024, with the same-day expiry. To calibrate the model, we estimate the risk-neutral parameters using equation (19) with option prices specified by equations (8) and (9). We set the instantaneous risk-free rate r_t and the continuously

compounded dividend yield δ_t to their respective annualized market-implied values for each day. The ex-post instantaneous variances V_t for each particular pricing time t were obtained separately from the Gibbs sampler as a by-product of recursive samplings from the filtering distributions (see subsection 4.2 for details).

The risk-neutral parameter estimates are given in the left panel of Table 6. The risk-neutral speed of mean reversion a^* is higher than its physical counterpart, while the corresponding long-run equilibrium level $\bar{V}^*$ is proportionally lower. The jump intensity parameters λ_0^* and λ_1^* are approximately 70 percent higher than their estimates under the physical measure, implying that investors price 0DTEs as if jumps are subjectively more likely, about 7–8 times per day. The risk-neutral expectation of the jump size k^* is also higher in absolute terms, with a value of -2.78 bp.

The right panel shows the corresponding risk premia, in basis points per day, calculated from the expressions in equation (7). These results align with the insight that intraday jumps have a significant and immediate impact on 0DTE option prices. The combined premium for the unknown jump arrival times and jump sizes is 12.21 basis points per day. This value is more than twice the return diffusion risk premium of 5.34 bp per day. The volatility risk premium, which captures the compensation for unpredictable intraday changes in volatility, is 1.28 bp per day, an order of magnitude smaller than the jump risk premium. Jumps thus contribute significantly more to the overall premium than the combined diffusion and volatility risk. Hence, investors interested in the ultra-short option tenors appear to be more concerned with intraday jumps than with the "regular" equity and volatility risks: the overall jump premium is almost double the combined diffusion and volatility risk premium.

Pearson's chi-square statistic of 450.36 indicates that the fit provided by equation (19)

Table 6: Risk-neutral parameters and risk premia implied from SPXW 0DTE option prices. The left panel of this table shows the values of risk-neutral parameters estimated via equation (19) on 22,410 observations of SPXW calls and puts in 2024. The right panel shows the corresponding risk premia, in basis points per day, calculated from equation (7). The annualized instantaneous risk-free rate r_t and the continuously compounded dividend yield δ_t are set to their respective market values for each day. The ex-post instantaneous variances V_t were obtained separately from the Gibbs sampler, as described in subsection 4.2. Standard errors of the estimates, shown in parentheses, were obtained by applying error propagation rules on their physical counterparts from Table 5.

Parameter	Value
a^*	0.1086
	(0.0744)
$\bar{V}^*$	19.2884
	(0.0397)
λ_0^*	0.0993
	(0.0453)
λ_1^*	31.1032
	(0.0000)
k^*	−2.7786
	(0.5703)
$\chi^2[22409]$	450.3632
p-value	1.0000

Premium	Value (bp/day)
Return diffusion risk	5.34
	(0.98)
Volatility risk	1.28
	(0.03)
Overall jump risk	12.21
	(5.28)

is highly accurate, i.e., that the model closely matches the observed SPXW data.[8] We can further visualize this in Figure 7, which compares the market and the model 0DTE option prices. The gray ovals, corresponding to the bid-ask spread for each of the 22,410 sample calls and puts, are concentrated around the parity line, where the model perfectly prices an option. If we "zoom in" on the graph segment containing most observations, the fit quality does not deteriorate.

[8] The chi-square critical value with 22,409 degrees of freedom at 0.05 significance is 22,061.92.

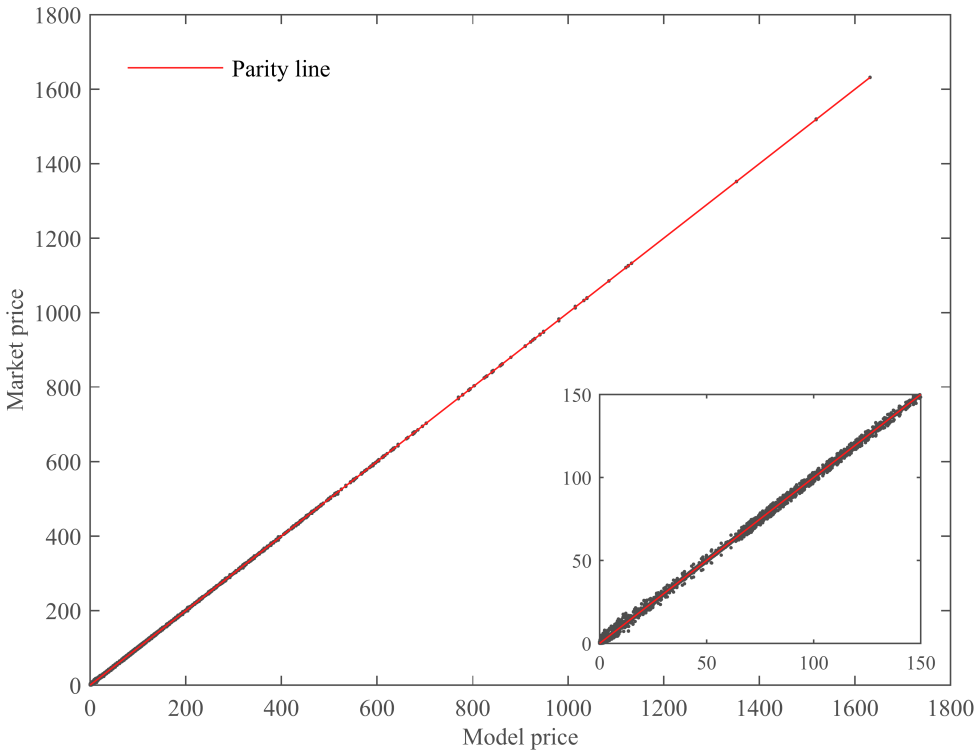


Figure 7: The semi-closed-form option-pricing model applied to SPXW data. This figure compares the market and the model 0DTE option prices. The gray ovals correspond to the bid-ask spreads for the 22,410 sample calls and puts with root SPXW during 2024. The corresponding values are on the vertical axis. They are plotted against the prices obtained from equations (8) and (9) while using the same underlying prices, strikes, residual maturities and types, and the full-model estimates of the structural parameters, $\widehat{\psi}$, from Table 5. The annualized instantaneous risk-free rate r_t and the continuously compounded dividend yield δ_t are set to their respective market values for each day. The ex-post instantaneous variances V_t were obtained separately from the Gibbs sampler, as described in subsection 4.2. The red line is the parity line, where the model perfectly prices an option. The inset graph shows details of the main graph, focusing on options traded at or below 150 points.

4.4 Practical hedging implications

Large intraday jumps in the SPX play a dominant role in shaping the risk premia of ultra-short-maturity options. These jumps are significant in both frequency and magnitude, with left-skewed return distributions. This asymmetry reflects an amplified sensitivity to downside tail risk that is strongly priced into 0DTEs throughout the trading day. Consequently, accurately anticipating the timing and severity of jumps is crucial for institutional risk management, particularly for liquidity provision and delta hedging.

To illustrate the sensitivity of 0DTEs to large intraday jumps and the limitations of conventional hedging models, we simulate a stylized example in which the SPX experiences a negative jump of 2.5 standard deviations at 10:30 a.m. Using risk-neutral parameters calibrated from market data (Tables 5 and 6), we generate 10,000 Monte Carlo paths under the SVJD-B model, each spanning 79 five-minute intervals. At each path, we initiate an at-the-money straddle, with a strike of $X = 5475.00$ and initial SPX level set to the June 18, 2024 value of 5476.15. For each simulated path, we compute the ex-ante option's delta and gamma immediately before the jump (i.e., at 10:25 a.m.), and its actual value change across the jump. The hedge error for the period between t and $t + \Delta t$ is then calculated using a second-order Taylor approximation:

$$\text{Hedge Error}_{t+\Delta t} = \text{P\&L from jump}_{t+\Delta t} - \left[\Delta_t \left(S_{t+\Delta t} - S_t\right) + \frac{1}{2}\Gamma_t \left(S_{t+\Delta t} - S_t\right)^2\right]. \quad (20)$$

The hedge error is defined as the difference between the actual change in the option's value and the hedge profit and loss (P&L), and is reported for both a jump-aware (SVJD-B) hedger and a Black-Scholes (BS) hedger. We also compute ex post delta and gamma, $\Delta_{t+\Delta t}$ and $\Gamma_{t+\Delta t}$, to quantify the distortion introduced by relying solely on ex ante estimates. The semi-closed form expressions for delta and gamma are given in Appendix D.

The simulation results, presented in Table 7, demonstrate the significant impact of model choice and jump risk on hedge ratios and P&L accuracy. Before the simulated jump, both the SVJD-B and BS models produce similar deltas and gammas. However, the SVJD-B model yields slightly lower sensitivities, reflecting its adjustment for latent jump risk. Nevertheless, the hedge errors prior to the jump are similar and positive under both models, suggesting that both hedges broadly capture the option's local risk exposure in the absence of jumps. After the jump, the differences become more pronounced. The ex-post delta under both models shifts sharply negative, consistent with the straddle becoming deep in-the-money on the put leg. Notably, the SVJD-B model produces a smaller absolute hedge error (-13.05) than BS (-56.68), indicating a substantial improvement in hedging precision when jumps are explicitly modeled. The standard errors further confirm the robustness of this difference.

Table 7: The impact of jumps on hedge ratios and P&L errors. This table reports ex-post delta and gamma hedge ratios, as well as hedge errors, for a 0DTE at-the-money straddle around a simulated SPX jump of 2.5 standard deviations occurring at 10:30 a.m. Hedge errors are defined by equation (20) as the difference between the option's actual value change across the jump and the P&L of a delta/gamma hedge constructed using the ex-ante hedge ratios. The hedge is implemented under either the full SVJD-B model or the Black-Scholes (BS) diffusion model. Results are based on 10,000 simulated paths, each with 79 five-minute intervals. We use the structural parameter estimates from Table 5, while the risk premia match those in Table 6. The initial SPX level is 5476.15, and the drift parameter is $r_t - \delta_t = 3.9480\%$, based on market data from June 18, 2024. The strike price of the 0DTE options is $X = 5475.00$. Standard errors from simulation are reported in parentheses.

	Before jump		After jump	
	SVJD-B	BS	SVJD-B	BS
$\Delta_{t+\Delta t}$	0.1062	0.1165	-0.5797	-0.6205
	(0.0051)	(0.0056)	(0.0041)	(0.0043)
$\Gamma_{t+\Delta t}$	0.0424	0.0460	0.0288	0.0299
	(0.0001)	(0.0002)	(0.0002)	(0.0002)
Hedge Error$_{t+\Delta t}$	32.1932	29.3917	-13.0548	-56.6782
	(0.3258)	(0.4701)	(2.2213)	(3.5470)

To better understand the sources of hedge error reported in Table 7, we illustrate how the pricing and sensitivities of a 0DTE straddle evolve throughout the trading day under both the SVJD-B and BS models (Figure 8). For simplicity, the underlying index level is set at $S_t = 1.01X$, and the instantaneous variance is fixed at its long-run mean, $V_t = \bar{V}$. These values are held constant to isolate the effect of model dynamics on option pricing and Greeks, excluding the impact of changing market inputs. The top-left panel shows that the SVJD-B model consistently assigns a higher premium to the straddle throughout the trading day, capturing additional compensation for jump and stochastic volatility risks that are absent in the BS model. As expiration nears, both models exhibit declining straddle values due to time decay. However, the SVJD-B premium remains elevated until late in the session.

The top-right panel presents the delta of the straddle, which remains below the BS benchmark under the SVJD-B model. This gap reflects the market's anticipation of potential negative jumps. With a moderately in-the-money call and out-of-the-money put, the SVJD-B model incorporates downside jump risk, reducing the effective directional sensitivity relative to BS. As time to maturity shortens, both deltas increase and converge toward 1, consistent with the dominance of intrinsic value near expiry.

The bottom-left panel shows that gamma under SVJD-B is persistently higher than under BS. There is a sharper change in delta for small movements in the underlying, originating from the model's responsiveness to discontinuous price moves. Higher gamma can be interpreted as compensation for the increased convexity risk borne by straddle sellers in environments with jump uncertainty.

Lastly, the bottom-right panel reveals a time-varying pattern in vega. The SVJD-B vega begins below the BS counterpart but crosses above it in the afternoon session. Early in the day, the jump component absorbs much of the risk, reducing sensitivity to changes

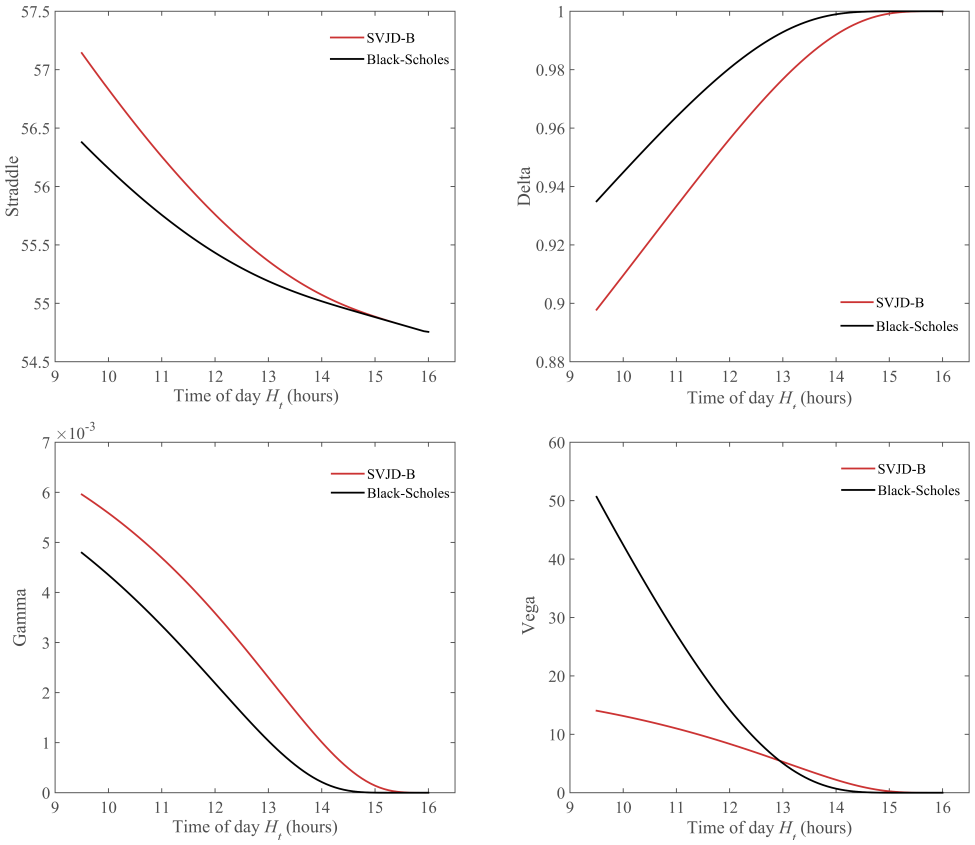


Figure 8: Intraday evolution of the value and Greeks of a 0DTE SPX straddle under two pricing models. The four panels show the straddle's price, delta, gamma and vega from market open (9:30) to close (16:00), assuming the SPX index remains constant at 1.01 of the option strike, and the instantaneous variance remains constant at its long-run mean. Results are plotted for the SVJD-B model (red) and the Black-Scholes model (black), highlighting the role of jumps and stochastic volatility in shaping intraday risk profiles.

in instantaneous variance. As the trading day progresses and jump uncertainty begins to resolve, the remaining exposure to volatility becomes more pronounced, leading to slightly elevated vega under SVJD-B near expiry.

5 Conclusion

This paper examines the pricing and hedging implications of intraday jumps on ultra-short-term index options. We develop a continuous-time stochastic volatility model with Poisson jumps of time-varying intensity to model the dynamics of the index. This setup allows us to derive the prices of the corresponding European call and put options in a semi-closed form. We use 9,164 SPX five-minute bars in the underlying index and infer the model parameters by minimizing the weighted score of the auxiliary model using the Efficient Method of Moments. The auxiliary model follows a semi-non-parametric specification and incorporates deterministic diurnal patterns in intraday data. For the corresponding 0DTE options, we use 22,410 five-minute bars with root SPXW to assess the market-implied risk premia.

Our results indicate that intraday jumps in the SPX are critical drivers of the 0DTE price dynamic. Firstly, they are significant in frequency, with around 4.5 daily jumps. Secondly, they are significant in magnitude as they move the index by about 2.5 standard deviations in five-minute returns on average. Negative jumps are slightly more recurring and impactful than positive ones. Jumps occur more frequently during high-volatility periods and tend to cluster near the market's opening and closing hours. The total jump-risk premium implied from 0DTE prices is almost double that for return-diffusion and volatility risk combined.

Methodologically, our proposed SVJD-B model is one of the rare studies to develop an option-pricing framework accurately applicable to 0DTEs, specifically designed to capture their complex intraday dynamics. Its setup, including a bimodal jump distribution and an affine structure (as discussed in Section 2), enables a semi-closed-form solution for ultra-short-term European options and their Greeks. This framework separates unpredictable return discontinuities from predictable diurnal volatility patterns, allowing for granular,

large-scale calibration across tens of thousands of 0DTE contracts and providing a more flexible fit to intraday skew and smile patterns. Empirically, the paper's key contribution is the finding that the jump risk premium is substantial and is undoubtedly priced by 0DTEs, almost double that for return-diffusion and volatility risk combined.

The results have significant implications for investors, hedgers, and risk managers. 0DTE options can serve as practical tools for managing exposure to extreme intraday price movements, which drive much of the pricing dynamics in these ultra-short-dated instruments. A simulation of a large SPX jump shows that while a pure-diffusion model produces hedge ratios broadly similar to those from a model incorporating stochastic volatility and jumps, the latter results in markedly smaller absolute hedging errors. Accounting for discontinuous price changes thus enables more accurate ex-ante risk assessments and more effective delta-gamma hedging. For investors seeking protection against sudden, event-driven market moves, 0DTEs offer more reliable risk mitigation when paired with jump-aware hedging strategies. At the same time, the sizeable hedge errors under the pure-diffusion model underscore the dangers of underestimating jump risk.

The findings further suggest that 0DTE options can also be used to speculate on extreme intraday market movements. Since the jump risk premium is both considerable and explicitly priced into 0DTEs, investors willing to bear it can use naked positions to express directional views on the likelihood of extreme price changes. The dual application of these options in hedging and speculation likely explains the explosive growth in demand for 0DTE options over the past two years.

The model's practical value stems from its ability to isolate and quantify intraday jump risk, which is an essential feature for instruments like 0DTE options, whose entire risk exposure is concentrated within a single trading day. Unlike slightly longer-term options (e.g., one-day or one-week) that smooth over risk across multiple trading sessions, 0DTEs are

highly sensitive to real-time tail events, such as macroeconomic announcements or volatility spikes near the market open or close. This feature makes them especially well-suited for targeted hedging of intraday tail risk. Our framework captures this microstructure by explicitly modeling time-varying jump intensities and jump-size asymmetry.

While this paper contributes to understanding 0DTEs and their pricing dynamics, it also faces several limitations that should be noted. There are at least three methodological issues with the proposed approach when applied to intraday data. The first one concerns the auxiliary model, where the polynomial expansion parameters in the semi-nonparametric specification are sensitive to the choice of initial values. This sensitivity may affect the precision with which the auxiliary score is estimated in the simulation step. The second issue concerns the computational intensity of the simulation step: we require densely sampled return paths to avoid discretization errors. This concern is partially resolved by pre-generating random numbers, so that iterations over the structural parameters do not require excessive repetition. Fortunately, the structural parameters must be estimated only once per sample of index returns. The third issue is computationally intensive risk-neutral parameter calibration, requiring tens of thousands of options to be repriced at each numerical optimizer iteration. For this point, it is crucial that our model yields a semi-closed-form option pricing formula and that only three free parameters need to be estimated. Usual code optimization and parallel processing techniques can further improve computational efficiency.

The key empirical limitation lies in the time coverage and the specific nature of our dataset. We purposefully selected the period from January to June 2024 because it proved exceptionally rich for observing intraday jumps and their profound impact on 0DTE options, reflecting dynamic monetary policy shifts, significant macroeconomic data releases and major earnings announcements. While our approach is robust enough to capture the

dynamics within such event-driven, volatile environments, this very specificity could, by design, limit the generalizability of our findings to other market conditions. However, the time coverage of the index data is less consequential, since they are primarily used in the auxiliary model and do not critically affect the simulation-estimated structural parameters. Nevertheless, extending both the SPX and SPXW 0DTE data to encompass longer and more diverse market regimes, including calmer periods and different economic cycles, would further enhance the external validity of our results and offer a more comprehensive understanding of jump risk across varying market conditions.

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Appendix

A The risk-neutral version of the model

This Appendix outlines the derivation of the risk-neutral version of the model. Given that the diffusion and the jump process are independent of each other, we can split the return dynamics into the pure-diffusion part and the pure-jump part:

$$\frac{\mathrm{d}S_t}{S_t} = \left(\frac{\mathrm{d}S_t}{S_t}\right)_{\text{diff}} + \mathrm{d}J_t, \tag{A1}$$

where

$$\left(\frac{\mathrm{d}S_t}{S_t}\right)_{\text{diff}} = \mu \mathrm{d}t + \sqrt{V_t}\, \mathrm{d}W_{1,t} \tag{A2}$$

and

$$\mathrm{d}J_t = (e^{u_t} - 1)\, \mathrm{d}q_t - \lambda_t k \mathrm{d}t. \tag{A3}$$

The pure-diffusion return given by equation (A2) and the instantaneous volatility V_t given by equation (2) follow a joint Brownian diffusion. Define

$$\mathrm{d}\mathbf{Y}_t = \sqrt{V_t} \begin{bmatrix} 1 & 0 \\ 0 & \theta \end{bmatrix} \mathrm{d}\mathbf{W}_t, \tag{A4}$$

for all t. To find the risk-neutral equivalent $\mathrm{d}\mathbf{Y}^*$ of (A4) that would be a martingale under an equivalent measure $\mathbb{P}^*$, we first write the Radon-Nikodým derivative of $\mathbb{P}^*$ with respect to the physical measure $\mathbb{P}$:

$$\frac{\mathrm{d}\mathbb{P}^*}{\mathrm{d}\mathbb{P}} = \exp\left[-\int_0^t \boldsymbol{\xi}_s \cdot \mathrm{d}\mathbf{Y}_s - \frac{1}{2}\int_0^t (\boldsymbol{\xi}_s \cdot \mathrm{d}\mathbf{Y}_s)\,(\mathrm{d}\mathbf{Y}_s \cdot \boldsymbol{\xi}_s)\right],$$

where $\boldsymbol{\xi}_s$ is a two-dimensional vector measurable with respect to $\mathcal{F}_s$ (see Bingham & Kiesel, 2004). Then, by Girsanov's theorem, a $\mathbb{P}^*$-Brownian motion has the form

$$\mathrm{d}\mathbf{Y}_t^* = \mathrm{d}\mathbf{Y}_t \left(1 + \mathrm{d}\mathbf{Y}_t \cdot \boldsymbol{\xi}_t\right).$$

Therefore,

$$\mathrm{d}\mathbf{Y}_t \;=\; \mathrm{d}\mathbf{Y}_t^* - \begin{bmatrix} 1 & \rho\theta \\ \rho\theta & \theta^2 \end{bmatrix} \begin{bmatrix} \xi_{1,t} \\ \xi_{2,t} \end{bmatrix} V_t \mathrm{d}t,$$

which implies that we can substitute

$$\begin{bmatrix} \mathrm{d}W_{1,t} \\ \mathrm{d}W_{2,t} \end{bmatrix} = \begin{bmatrix} \mathrm{d}W_{1,t}^* - \left(\xi_{1,t} + \rho\theta\xi_{2,t}\right)\sqrt{V_t}\mathrm{d}t \\ \mathrm{d}W_{2,t}^* - \left(\rho\xi_{1,t} + \theta\xi_{2,t}\right)\sqrt{V_t}\mathrm{d}t \end{bmatrix}$$

into (1) and (2). Hence, the processes

$$\left(\frac{\mathrm{d}S_t}{S_t}\right)_{\text{diff}} = (r_t - \delta_t)\mathrm{d}t + \sqrt{V_t}\;\mathrm{d}W_{1,t}^*$$

and

$$\mathrm{d}V_t = a^*\left(\bar{V}^* - V_t\right)\mathrm{d}t + \theta\sqrt{V_t}\;\mathrm{d}W_{2,t}^*$$

both contain diffusions that are (jointly) martingales under $\mathbb{P}^*$, as long as

$$\begin{aligned} \xi_{1,t} + \rho\theta\xi_{2,t} &= \frac{\mu - (r_t - \delta_t)}{V_t}, \\ \theta\left(\rho\xi_{1,t} + \theta\xi_{2,t}\right) &= a^* - a, \\ \bar{V}^* &= \frac{a}{a^*}\bar{V}. \end{aligned}$$

The jump component in equation (A3) is a $\mathbb{P}$-martingale by construction:

$$\begin{aligned}\mathbb{E}_t(\mathrm{d}J_t) &= \mathbb{E}_t\left[(e^{u_t}-1)\mathrm{d}q_t\right] - \lambda_t k \mathrm{d}t \\ &= \mathbb{E}_t(e^{u_t}-1)\lambda_t \mathrm{d}t - \lambda_t k \mathrm{d}t \\ &= 0.\end{aligned}$$

The second equality follows from the $\mathcal{F}_t$-measurability of V_t. Define

$$\mathrm{d}N_t = \mathrm{d}q_t - \lambda_t \mathrm{d}t.$$

By applying Girsanov's theorem for point processes (Elliott & Kopp, 2005), the risk-neutral version of this process will be

$$\begin{aligned}\mathrm{d}N_t^* &= \mathrm{d}N_t - \mathbb{E}_t\left[\frac{e^{\xi_3+\xi_4 u_t}}{\mathbb{E}_t(e^{\xi_4 u_t})} - 1\right]\lambda_t \mathrm{d}t \\ &= \mathrm{d}N_t - (e^{\xi_3}-1)\lambda_t \mathrm{d}t \\ &= \mathrm{d}q_t - \lambda_t^* \mathrm{d}t,\end{aligned}$$

where the market prices of jump risk ξ_3 and ξ_4 are measurable with respect to $\mathcal{F}_t$, and $\lambda_t^* \equiv e^{\xi_3}\lambda_t$. Girsanov's theorem applied to $\mathrm{d}J$ then yields

$$\begin{aligned}\mathrm{d}J_t^* &= \mathrm{d}J_t - \mathbb{E}_t\left[\left(e^{\xi_3}\frac{e^{\xi_4 u_t}}{\mathbb{E}_t(e^{\xi_4 u_t})} - 1\right)(e^{u_t}-1)\right]\lambda_t \mathrm{d}t \\ &= (e^{u_t}-1)\mathrm{d}q_t - \lambda_t^*\left[\frac{Q(\xi_4+1)}{Q(\xi_4)} - 1\right]\mathrm{d}t,\end{aligned}$$

where

$$Q(\Phi) \equiv \mathbb{E}_t(e^{\Phi u_t}) = p(1+k_+)^{\Phi} e^{\frac{1}{2}\Phi(\Phi-1)\sigma_+^2} + (1-p)(1+k_-)^{\Phi} e^{\frac{1}{2}\Phi(\Phi-1)\sigma_-^2}.$$

Therefore, the process

$$\mathrm{d}J^* = (e^{u_t} - 1)\mathrm{d}q_t - \lambda_t^* k^* \mathrm{d}t$$

will be a martingale under $\mathbb{P}^*$ as long as

$$k^* = \frac{Q(\xi_4 + 1)}{Q(\xi_4)} - 1.$$

The parameter ξ_3 captures the market's inability to time the arrival of jumps, while ξ_4 measures the uncertainty related to the jump size and, possibly, the model uncertainty. Liu, Pan, and Wang (2004) also argue that a significant part of the jump risk premium should come from the uncertainty aversion in the sense of Knight (1921) and Ellsberg (1961).

Putting everything together, the processes

$$\begin{aligned} \frac{\mathrm{d}S_t}{S_t} &= (r_t - \delta_t)\mathrm{d}t + \sqrt{V_t}\mathrm{d}W_{1,t}^* + (e^{u_t} - 1)\,\mathrm{d}q_t - \lambda_t^* k^* \mathrm{d}t, \\ \mathrm{d}V_t &= a^*\left(\bar{V}^* - V_t\right)\mathrm{d}t + \theta\sqrt{V_t}\mathrm{d}W_{2,t}^*, \end{aligned}$$

with $\mathrm{d}W_{1,t}^*\mathrm{d}W_{2,t}^* = \rho\mathrm{d}t$, represent the risk-neutral equivalents of (1) and (2). The market risk premia are given by:

$$\begin{aligned} \text{return diffusion risk premium} &= \mathbb{E}\left[\mu - (r_t - \delta_t)\right] = \mu - (\bar{r} - \bar{\delta}), \\ \text{volatility risk premium} &= \mathbb{E}\left[a\left(\bar{V} - V_t\right) - a^*\left(\bar{V}^* - V_t\right)\right] = -(a - a^*)\bar{V}, \\ \text{overall jump risk premium} &= \mathbb{E}\left(\lambda_t k - \lambda_t^* k^*\right) = (\lambda_0 + \lambda_1\bar{V})k - (\lambda_0^* + \lambda_1^*\bar{V}^*)k^*. \end{aligned}$$

B Semi-closed-form solution for option prices

This Appendix outlines the derivation of a semi-closed-form solution for the option prices. Under the risk-adjusted model (5)–(6), the price at t of a European call option with residual maturity $\tau = T - t$ and strike price X is given by

$$\begin{aligned} C_t(S_t, V_t, \tau, X; \widehat{\psi}) &= \mathbb{E}_t^* \left[e^{-r_t(T-t)} \max\left(S_T - X, 0\right) \right] \\ &= e^{-r_t\tau} \mathbb{E}_t^* \left(S_T \mid S_T > X \right) - e^{-r_t\tau} X \mathbb{E}_t^* \left(1 \mid S_T > X \right) \\ &= e^{-\delta_t\tau} S_t \Pi_1 - e^{-r_t\tau} X \Pi_2, \end{aligned} \tag{B1}$$

where $\mathbb{E}_t^*(\cdot)$ denotes the expectation with respect to the risk-neutral probability measure $\mathbb{P}^*$ and conditional on the sigma-algebra $\mathcal{F}_t$. Likewise, the price at t of a European put option with residual maturity $\tau = T - t$ and strike price X is given by

$$\begin{aligned} P_t(S_t, V_t, \tau, X; \widehat{\psi}) &= \mathbb{E}_t^* \left[e^{-r_t(T-t)} \max\left(X - S_T, 0\right) \right] \\ &= e^{-r_t\tau} X \mathbb{E}_t^* \left(1 \mid S_T < X \right) e^{-r_t\tau} \mathbb{E}_t^* \left(S_T \mid S_T < X \right) \\ &= e^{-r_t\tau} X \left(1 - \Pi_2\right) - e^{-\delta_t\tau} S_t \left(1 - \Pi_1\right), \end{aligned} \tag{B2}$$

which also follows trivially from the put-call parity. The function Π_1 in equations (B1) and (B2) can be interpreted as the expected value of the underlying asset price conditionally on the call option being in the money. Similarly, Π_2 can be interpreted as the probability of a call being in the money.

The semi-closed-form expressions for Π_1 and Π_2 can be obtained via Fourier inversion of the complex-valued moment-generating functions F_j:

$$\Pi_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\mathrm{Im}\left[F_j(\mathrm{i}\Phi) e^{-\mathrm{i}\Phi x} \right]}{\Phi} \mathrm{d}\Phi, \tag{B3}$$

for $j = 1, 2$, where $x \equiv \ln(X/S_t)$. We can relate F_1 to the price of a contingent claim that pays S_T at time T, conditionally on $S_T > X$. Similarly, we can relate F_2 to the price of a contingent claim that pays $\exp(\Phi \ln S_T)$ at time T. Following a procedure analogous to the one presented in Bates (1996), we apply Itô's lemma to these prices and obtain the moment-generating functions as solutions to two separate integro-differential equations. We then seek the solutions to these equations in the following general form:

$$F_j(\Phi; V_t, \tau) = \exp\left[A_j(\tau; \Phi) + B_j(\tau; \Phi)V_t + \lambda_0^* \tau C_j(\Phi)\right], \tag{B4}$$

for $j = 1, 2$. The functions A_j, B_j and C_j solve a set of ordinary differential equations. They are given by:

$$\begin{aligned}
A_j(\tau; \Phi) &= (r_t - \delta_t)\tau\Phi - \frac{a\bar{V}\tau}{\theta^2}\left(\rho\theta\Phi - a_j - g_j\right) \\
&\quad - \frac{2a\bar{V}}{\theta^2}\ln\left[1 + \left(\rho\theta\Phi - a_j - g_j\right)\frac{1 - e^{g_j\tau}}{2g_j}\right], \qquad \text{(B5)} \\
B_j(\tau; \Phi) &= -\frac{\Phi^2 + (3 - 2j)\Phi + 2\lambda_1^* C_j(\Phi)}{\rho\theta\Phi - a_j + g_j(1 + e^{g_j\tau})/(1 - e^{g_j\tau})}, \qquad \text{(B6)} \\
C_j(\Phi) &= Q^*(\Phi + 2 - j) - Q^*(2 - j) - k^*\Phi, \qquad \text{(B7)}
\end{aligned}$$

where

$$\begin{aligned}
a_j &= a^* + (j - 2)\rho\theta, \\
g_j &= \sqrt{(\rho\theta\Phi - a_j)^2 - \theta^2\left[\Phi^2 + (3 - 2j)\Phi + 2\lambda_1^* C_j(\Phi)\right]}, \\
Q^*(\Phi) &\equiv \mathbb{E}_t^*(e^{\Phi u_t}) = \mathbb{E}_t\left[\frac{e^{\xi_4 u_t}}{\mathbb{E}_t(e^{\xi_4 u_t})}e^{\Phi u_t}\right] = \frac{Q(\xi_4 + \Phi)}{Q(\xi_4)} = (1 + k^*)\frac{Q(\xi_4 + \Phi)}{Q(\xi_4 + 1)}, \\
Q(\Phi) &= p(1 + k_+)^\Phi e^{\frac{1}{2}\Phi(\Phi-1)\sigma_+^2} + (1 - p)(1 + k_-)^\Phi e^{\frac{1}{2}\Phi(\Phi-1)\sigma_-^2},
\end{aligned}$$

for $j = 1, 2$. By setting $p = 1$ and $\lambda_1^* \equiv 0$, we obtain the option pricing formula given in

Bates (1996) for currency options or in Bates (2000) for options on a stock market index.

Given τ, X, r_t, δ_t, S_t and V_t, we calculate the option prices C_t or P_t in equations (B1) or (B2) by substituting the expressions for A_j, B_j and C_j from equations (B5)–(B7) into (B4), and then computing the integral in equation (B3) numerically via adaptive Gauss-Kronrod quadrature.

C Gradient in analytical form

The expectation of the score function given by equation (16) sums the components of the auxiliary log-likelihood gradient. In this Appendix, we show the analytical form of these components. They are obtained by direct calculation of partial derivatives of $\ln f(y_t|\mathcal{F}_{t-1};\varphi)$. From equations (14) and (10), we have:

$$\ln f(y_t|\mathcal{F}_{t-1};\varphi) = -\frac{1}{2}\left(\sum_{i=0}^{2} b_i H_t^i + \ln h_t + z_t^2\right) + 2\ln P_K(z_t) - \ln \mathcal{I}(\mathbf{c}), \tag{C1}$$

where $\mathbf{c} = (c_1, c_2, \ldots, c_K)'$ and

$$\mathcal{I}(\mathbf{c}) \equiv \int_{-\infty}^{\infty} P_K^2(z) e^{-z^2/2} \mathrm{d}z. \tag{C2}$$

Using the relationship between z_t and $y_t = \bar{y} + \varepsilon_t$, s_t and h_t, it is straightforward to verify that:

$$\begin{aligned}
\frac{\partial \ln f}{\partial \bar{y}} &= \frac{1}{s_t\sqrt{h_t}}\left(z_t - \frac{2P'(z_t)}{P(z_t)}\right) + \frac{1}{2h_t}\left(z_t^2 - 1 - \frac{2z_tP'(z_t)}{P(z_t)}\right)\frac{\partial h_t}{\partial \bar{y}}, \\
\frac{\partial \ln f}{\partial b_i} &= \frac{1}{2}\left(z_t^2 - 1 - \frac{2z_tP'(z_t)}{P(z_t)}\right)\left(H_t^i + \frac{1}{h_t}\frac{\partial h_t}{\partial b_i}\right), \quad \text{for } i = 0, 1, 2, \\
\frac{\partial \ln f}{\partial \boldsymbol{\phi}} &= \frac{1}{2h_t}\left(z_t^2 - 1 - \frac{2z_tP'(z_t)}{P(z_t)}\right)\frac{\partial h_t}{\partial \boldsymbol{\phi}}, \\
\frac{\partial \ln f}{\partial c_i} &= 2\left(\frac{z_t^i}{P(z_t)} - \frac{\mathcal{J}_i(\mathbf{c})}{\mathcal{I}(\mathbf{c})}\right), \quad \text{for } i = 1, 2, \ldots, K,
\end{aligned}$$

where

$$\begin{aligned}
\boldsymbol{\phi} &= [\omega\ \alpha_1\ \ldots\ \alpha_{L_1}\ \beta_1\ \ldots\ \beta_{L_2}\ \gamma_1\ \ldots\ \gamma_{L_1}]' \\
\mathcal{J}_i(\mathbf{c}) &\equiv \int_{-\infty}^{\infty} z^i P_K(z) e^{-z^2/2} \mathrm{d}z \\
&= \sum_{j=0}^{K} c_j 2^{(i+j-1)/2} \left((-1)^{i+j}+1\right) \Gamma\left(\frac{i+j+1}{2}\right),
\end{aligned}$$

for $i = 1, 2, \ldots, K$, and Γ denotes the gamma function. The partial derivatives of h_t are computed recursively, using equation (11):

$$\begin{aligned}
\frac{\partial h_t}{\partial \bar{y}} &= -2\sum_{l=1}^{L_1}\left(\alpha_l + \gamma_l \mathbf{1}_{\{\eta_{t-l}<0\}}\right)\eta_{t-l} + \sum_{l=1}^{L_2}\beta_l \frac{\partial h_{t-l}}{\partial \bar{y}}, \\
\frac{\partial h_t}{\partial b_i} &= -\sum_{l=1}^{L_1}\left(\alpha_l + \gamma_l \mathbf{1}_{\{\eta_{t-l}<0\}}\right)\eta_{t-l}^2 H_{t-l}^i + \sum_{l=1}^{L_2}\beta_l \frac{\partial h_{t-l}}{\partial b_i}, \\
\frac{\partial h_t}{\partial \boldsymbol{\phi}} &= \mathbf{v}_t + \sum_{l=1}^{L_2}\beta_l \frac{\partial h_{t-l}}{\partial \boldsymbol{\phi}},
\end{aligned}$$

where

$$\mathbf{v}_t \equiv \left[1\ \ \eta_{t-L_1}\ \ldots\ \eta_{t-1}\ \ h_{t-L_2}\ \ldots\ h_{t-1}\ \ \eta_{t-L_1}\mathbf{1}_{\{\eta_{t-L_1}<0\}}\ \ldots\ \eta_{t-1}\mathbf{1}_{\{\eta_{t-1}<0\}}\right]'.$$

As the initial conditions, we use

$$\begin{aligned}
\frac{\partial h_t}{\partial \bar{y}} &= 0 \\
\frac{\partial h_t}{\partial b_i} &= 0 \quad \text{for } i = 0, 1, 2 \\
\frac{\partial h_t}{\partial \boldsymbol{\phi}} &= [1\ 0\ \ldots\ 0\ 0\ \ldots\ 0\ 0\ \ldots\ 0]' \\
\mathbf{v}_t &= [1\ \ h_0\ \ldots\ h_0\ \ h_0\ \ldots\ h_0\ \ h_0/2\ \ldots\ h_0/2]',
\end{aligned}$$

for $t = 0, 1, \ldots, L_2$, where

$$h_0 \equiv \frac{\omega}{1 - \sum_{l=1}^{L_1} (\alpha_l + \gamma_l/2) - \sum_{l=1}^{L_2} \beta_l}.$$

The initial conditions have a negligible effect on the results when the burn-in period is discarded.

D Option Greeks

The option Greeks follow directly from the option prices in the semi-closed form, given by equations (B1) and (B2). For example, delta (Δ), gamma (Γ) and vega ($\mathcal{V}$) for calls (C) and puts (P) can be obtained from a straightforward calculation of partial derivatives of option prices:

$$\Delta_t^C \equiv \frac{\partial C_t}{\partial S_t} = e^{-r_t\tau}\frac{X}{S_t}\mathfrak{D}_2^{(1)} + e^{-\delta_t\tau}\left(\Pi_1 - \mathfrak{D}_1^{(1)}\right), \tag{D1}$$

$$\Delta_t^P \equiv \frac{\partial P_t}{\partial S_t} = e^{-r_t\tau}\frac{X}{S_t}\mathfrak{D}_2^{(1)} - e^{-\delta_t\tau}\left(1 - \Pi_1 + \mathfrak{D}_1^{(1)}\right), \tag{D2}$$

$$\Gamma_t^C \equiv \frac{\partial^2 C_t}{\partial S_t^2} = \frac{1}{S_t}e^{-\delta_t\tau}\left(\mathfrak{D}_1^{(2)} - \mathfrak{D}_1^{(1)}\right) - \frac{X}{S_t^2}e^{-r_t\tau}\left(\mathfrak{D}_2^{(1)} + \mathfrak{D}_2^{(2)}\right), \tag{D3}$$

$$\Gamma_t^P \equiv \frac{\partial^2 P_t}{\partial S_t^2} = \frac{1}{S_t}e^{-\delta_t\tau}\left(\mathfrak{D}_1^{(2)} - \mathfrak{D}_1^{(1)}\right) - \frac{X}{S_t^2}e^{-r_t\tau}\left(\mathfrak{D}_2^{(1)} + \mathfrak{D}_2^{(2)}\right) = \Gamma_t^C, \tag{D4}$$

$$\mathcal{V}_t^C \equiv \frac{\partial C_t}{\partial \varsigma_t} = 2\varsigma_t\left(e^{-\delta_t\tau}S_t\mathfrak{V}_1 - e^{-r_t\tau}X\mathfrak{V}_2\right), \tag{D5}$$

$$\mathcal{V}_t^P \equiv \frac{\partial P_t}{\partial \varsigma_t} = 2\varsigma_t\left(e^{-\delta_t\tau}S_t\mathfrak{V}_1 - e^{-r_t\tau}X\mathfrak{V}_2\right) = \mathcal{V}_t^C. \tag{D6}$$

In equations (D1)–(D4),

$$\mathfrak{D}_j^{(n)} = \frac{1}{\pi}\int_0^\infty \frac{\mathrm{Im}\left[F_j(\mathrm{i}\Phi)e^{-\mathrm{i}\Phi x}(-\mathrm{i}\Phi)^n\right]}{\Phi}\mathrm{d}\Phi, \tag{D7}$$

for $j = 1, 2$ and $n = 1, 2$. In equations (D5) and (D4), $\varsigma_t \equiv \sqrt{V_t}$ is the instantaneous volatility, and

$$\mathfrak{V}_j = \frac{1}{\pi} \int_0^\infty \frac{\text{Im}\left[B_j(\tau; \Phi) e^{-\text{i}\Phi x}\right]}{\Phi} \text{d}\Phi, \tag{D8}$$

for $j = 1, 2$. All remaining quantities in equations (D1)–(D8) have the same meaning as in Appendix B.

The value of a straddle portfolio at any moment t is $C_t + P_t$. Thus, delta, gamma and vega for a straddle are given by:

$$\Delta_t = \Delta_t^C + \Delta_t^P, \tag{D9}$$

$$\Gamma_t = \Gamma_t^C + \Gamma_t^P, \tag{D10}$$

$$\mathcal{V}_t = \mathcal{V}_t^C + \mathcal{V}_t^P, \tag{D11}$$